



The Universal Business Operating System

Project Proposal Documentation (PPD)

Field	Detail
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"Connect Everything. Understand Everything. Automate Everything. Grow Everything."



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1. Cover Page

Horizon 360 CRM is presented here not as another customer relationship management tool, but as a category redefinition: a **Universal Business Operating System (UBOS)** that collapses the entire enterprise software estate — CRM, ERP, Finance, HRMS, Marketing, Sales, Projects, Service, Commerce, Vendor, Partner, Analytics, AI, and Workflow Automation — into a single, intelligent, AI-native platform built on one unified data model.

The platform is organized around an **8-Level Business Operating System**, a **BIOM Ecosystem** of composable business modules, an **AI Intelligence Mesh**, a durable **Workflow Engine**, an **Executive Intelligence Center**, an open **Integration Platform**, and a shared **Universal Data Model**.

 **Positioning Statement:** Where competitors sell *suites* of loosely-integrated products, Horizon 360 ships a *single organism* — every module breathes the same data, the same identity graph, and the same intelligence layer.

2. Executive Summary

2.1 Overview

Modern enterprises run on a patchwork of 80–130 disconnected SaaS applications. Each tool solves one problem and creates three more: duplicated records, brittle integrations, fractured analytics, and an AI layer that can never see the whole picture. **Horizon 360 CRM** resolves this by re-architecting business software around a single principle — *one data model, many capabilities, one intelligence*.

Horizon 360 is delivered as a multi-tenant, cloud-native platform composed of independently deployable **Business Intelligent Operating Modules (BIOMs)**. Each BIOM (Sales, Marketing, Finance, HRMS, Commerce, Service, Project, Vendor, Partner) is a first-class domain that shares the platform's Universal Data Model, identity fabric, workflow engine, and AI Intelligence Mesh.

2.2 Business Motivation

Driver	Description
 Cost Consolidation	Enterprises pay for CRM + ERP + HRMS + Marketing + BI separately, with overlapping seats and integration middleware. Horizon 360 consolidates spend.
 Data Unification	A single customer, vendor, employee, and transaction graph eliminates reconciliation overhead.

Driver	Description
 AI Leverage	AI is only as good as the data it sees. A unified model lets AI reason across the <i>entire</i> business, not one silo.
 Operational Velocity	Native workflow automation replaces manual hand-offs between disconnected tools.

2.3 Enterprise Value

Horizon 360 targets measurable outcomes: a **30–50% reduction in total software TCO**, **60–80% reduction in manual cross-system data entry**, and **single-pane executive visibility** across every function in real time. Because intelligence is native rather than bolted-on, decision latency — the time from event to insight to action — collapses from days to minutes.

2.4 Innovation

The defining innovations are: (1) the **Universal Data Model (UDM)** that treats every business object as a node in one graph; (2) the **AI Intelligence Mesh** that gives every module a domain-specialized agent sharing a common memory; (3) the **BIOM composability model** that lets customers activate only the modules they need without losing unification; and (4) a **durable Workflow Engine** (Temporal) that guarantees long-running business processes survive failures.

2.5 Market Opportunity

The global CRM market exceeds USD 90B, ERP exceeds USD 55B, and the broader business-applications software market is well over USD 300B annually — yet it remains fragmented across point solutions. Horizon 360 attacks the **convergence opportunity**: mid-market and enterprise organizations actively seeking to *reduce vendor sprawl*. (Market sizing figures here are indicative and should be re-validated against current analyst reports before external circulation.)

2.6 Expected Outcome

A production-grade, multi-tenant platform with nine activated BIOMs, an AI Intelligence Mesh, and an Executive Intelligence Center — capable of serving from a 20-person startup to a 50,000-seat enterprise on the same codebase, differentiated only by configuration and tenancy.

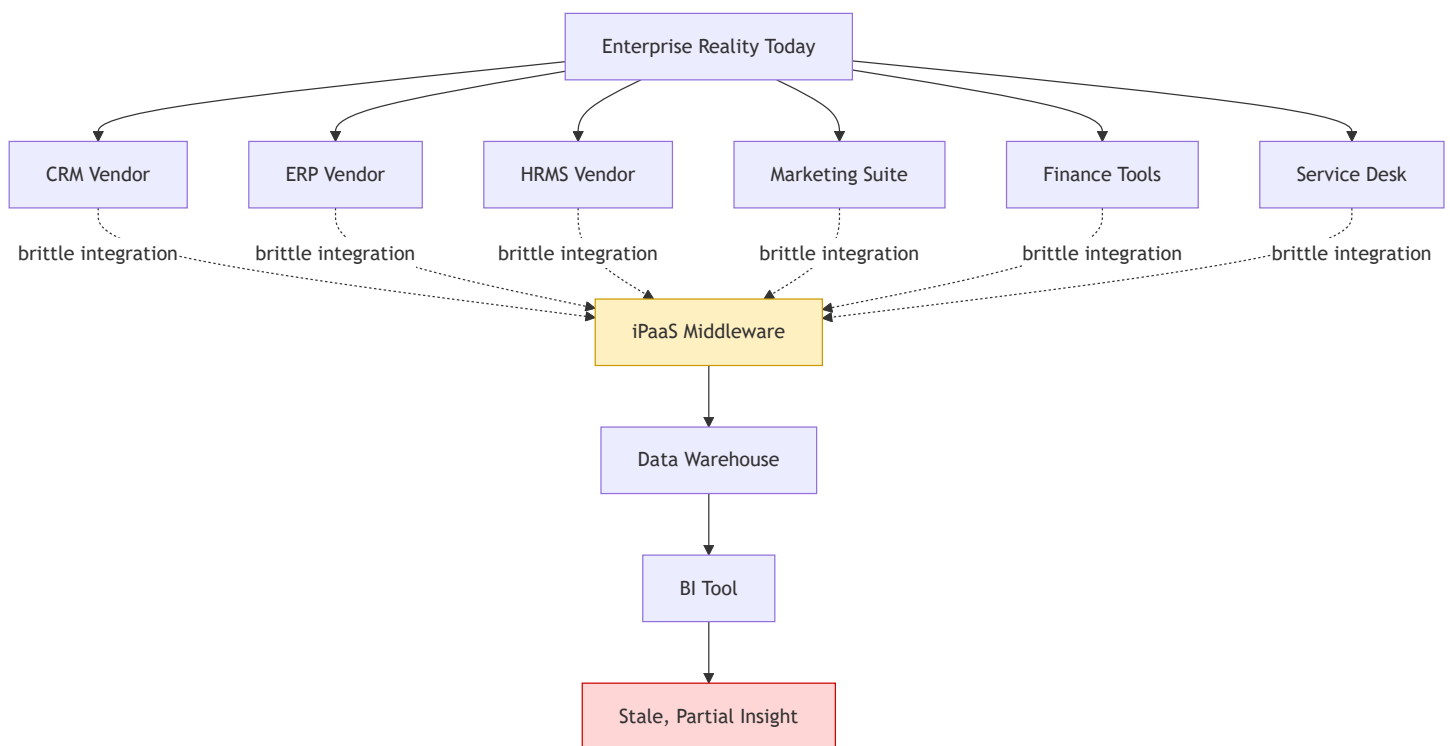
 **Best Practice Note:** All quantitative claims (TCO reduction, market size, ROI) in this document are modeling assumptions for proposal purposes. Per due-diligence standards, validate each against at least two independent primary sources before investor distribution.

3. Introduction

3.1 Background

Business software evolved as a series of point solutions. Salesforce defined cloud CRM. Workday defined cloud HR. NetSuite defined cloud ERP. Each was revolutionary in isolation — and each became an island. The integration tax that enterprises now pay (iPaaS tools, ETL jobs, reverse-ETL, data warehouses, reconciliation teams) frequently exceeds the cost of the applications themselves.

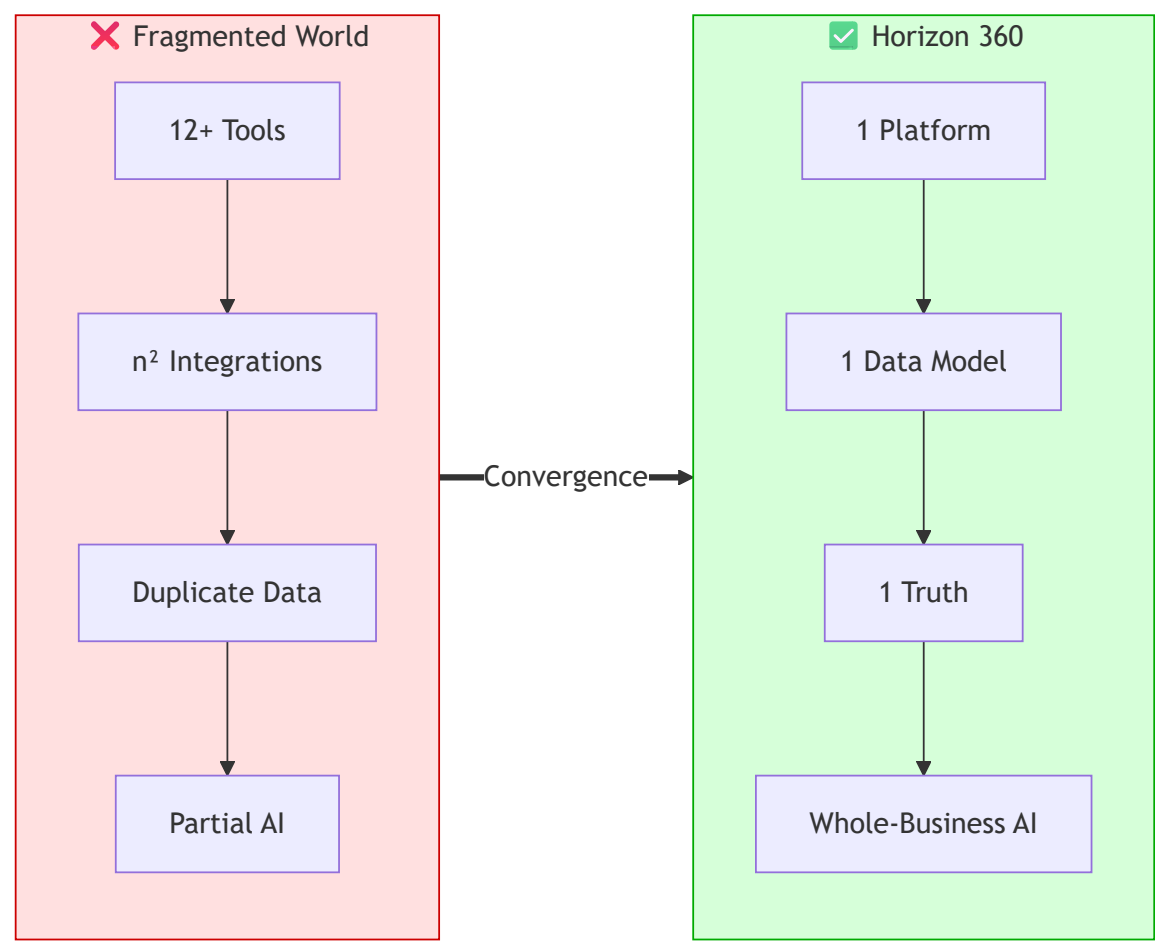
3.2 The Current Enterprise Software Landscape



3.3 Problems with Fragmented Business Tools

Fragmentation imposes a compounding cost. Every new tool multiplies the number of integration points (an n-squared problem), duplicates master data, and fractures the analytical layer. A "customer" exists differently in the CRM, the billing system, the support desk, and the marketing platform — and no system holds the truth.

3.4 The Need for a Unified Platform



The thesis of Horizon 360 is simple: the integration problem is unsolvable at the seams. It can only be solved at the **core** — by building every capability on one shared model from day one.

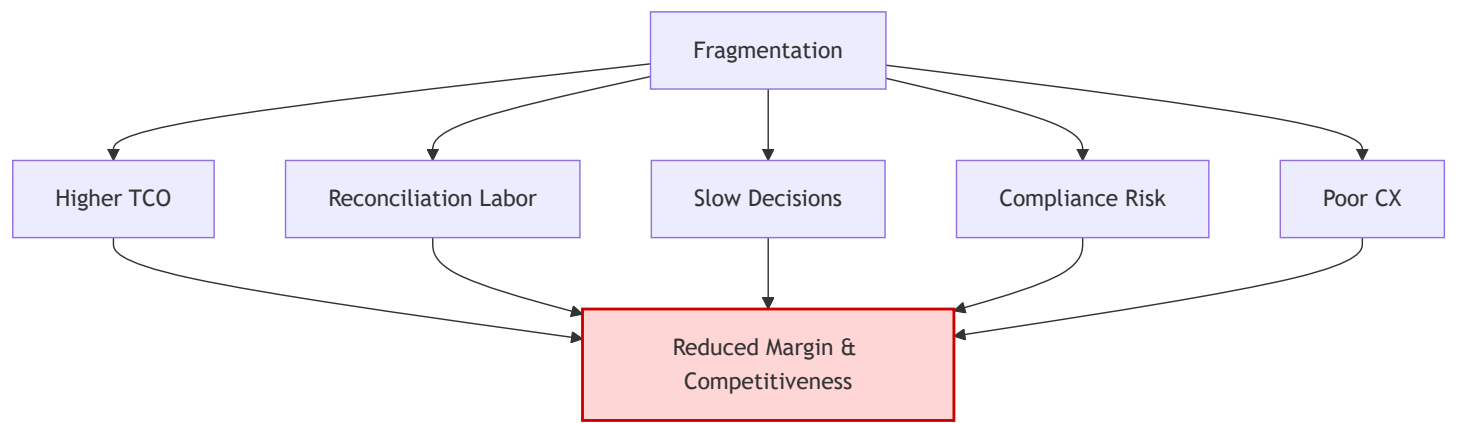
4. Problem Statement

4.1 Current Challenges

Challenge	Description	Business Impact
🔌 Disconnected CRMs	Sales, marketing, and service each maintain separate customer records.	Inconsistent customer experience; lost revenue.
💳 Multiple Subscriptions	10–15 overlapping SaaS contracts with redundant seats.	Inflated, unpredictable software spend.
📄 Data Duplication	The same entity exists in many systems with no master.	Reconciliation labor; reporting errors.

Challenge	Description	Business Impact
 Poor Analytics	BI is downstream, batch, and partial.	Decisions made on stale data.
 Lack of Automation	Manual hand-offs bridge system gaps.	Slow cycle times; human error.
 Limited AI	AI sees one silo and cannot reason across the business.	Shallow, untrustworthy intelligence.
 Manual Workflows	Approvals and processes run over email and spreadsheets.	No auditability; bottlenecks.

4.2 The Business Impact



 **Important Highlight:** The core problem is not "we need a better CRM." It is "every system holds a fraction of the truth, and no system holds all of it." Horizon 360 is built to *be* the truth.

5. Proposed Solution

5.1 Horizon 360 CRM — The Universal Business Operating System

Horizon 360 reframes the category. It is not a CRM with add-ons; it is a **Business Operating System** — a foundational layer on which all business functions execute, exactly as a computer OS is the layer on which all applications run.

8-Level Business Operating System

L1 · Universal Data Model



L2 · Identity & Tenancy
Fabric



L3 · Core Services Layer



L4 · BIOM Ecosystem



L5 · Workflow Engine



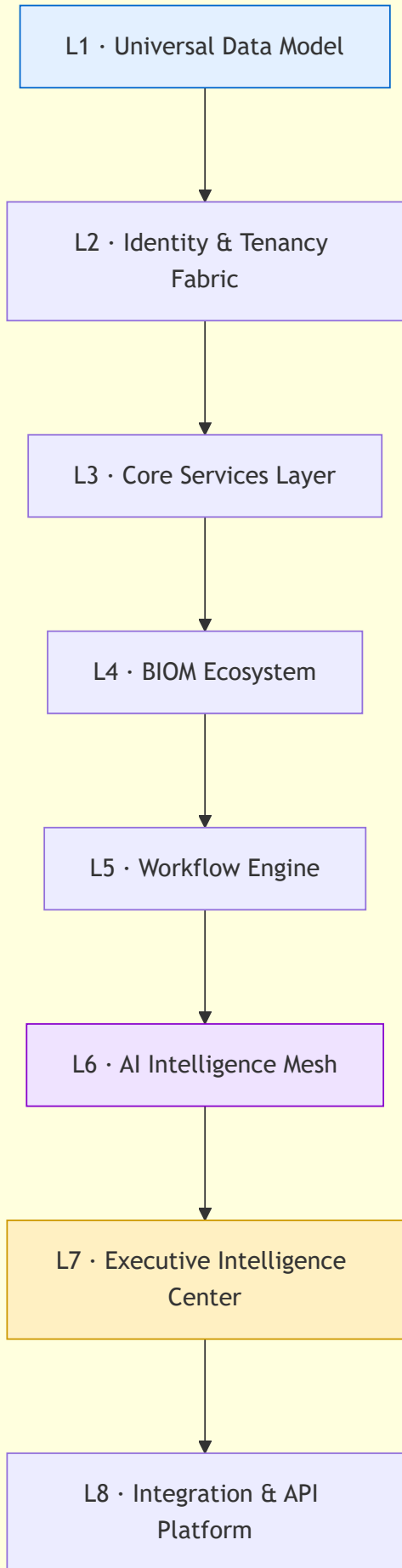
L6 · AI Intelligence Mesh



L7 · Executive Intelligence
Center



L8 · Integration & API
Platform



5.2 Single Source of Truth

Every business object — Account, Contact, Lead, Opportunity, Invoice, Employee, Vendor, Ticket, Project — is a node in one Universal Data Model. There is exactly one record for each real-world entity, referenced by every module.

5.3 Unified Customer Intelligence

Because Sales, Marketing, Service, Finance, and Commerce all read and write the same customer node, the platform holds a complete 360° view: every email, every invoice, every support ticket, every order, every campaign touch — unified, in real time.

5.4 AI-First Architecture

AI is not a feature bolted onto modules; it is **Level 6** of the operating system. The AI Intelligence Mesh gives every BIOM a specialized agent, all sharing common memory, retrieval, and guardrails.

5.5 Workflow Automation & Enterprise Scalability

A durable workflow engine (Temporal) orchestrates long-running, fault-tolerant business processes. Multi-tenancy (Stancl) and a microservice topology allow the same platform to scale from a single small business to a global enterprise.

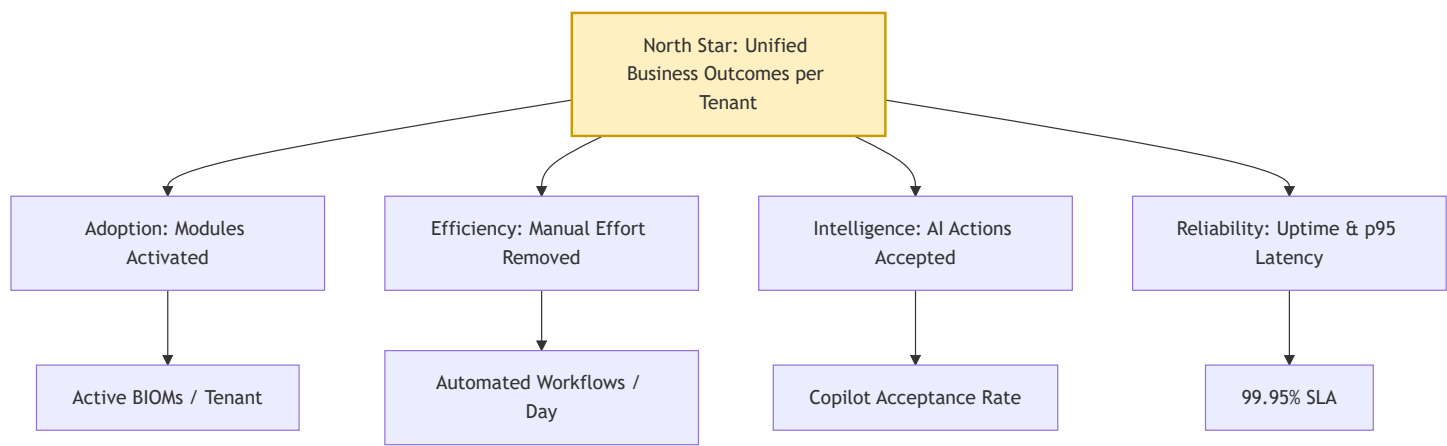
6. Product Objectives

6.1 Objective Matrix

Category	Objective	Success Metric (KPI)
 Strategic	Become the system of record for the entire enterprise.	% of business functions on platform
 Operational	Eliminate cross-system manual data entry.	60–80% reduction in manual entry
 Technical	Sub-300ms p95 API latency at 10k concurrent tenants.	p95 latency, tenant density

Category	Objective	Success Metric (KPI)
 Business	Reduce customer software TCO by 30–50%.	Consolidated spend per seat
 AI	Whole-business reasoning with grounded answers.	Answer groundedness / deflection rate
 Analytics	Real-time executive visibility across all BIOMs.	Dashboard freshness (seconds)

6.2 KPI Tree



 **Best Practice Note:** Each objective is paired with a measurable KPI. Objectives without metrics are aspirations; objectives with metrics are commitments.

7. Scope

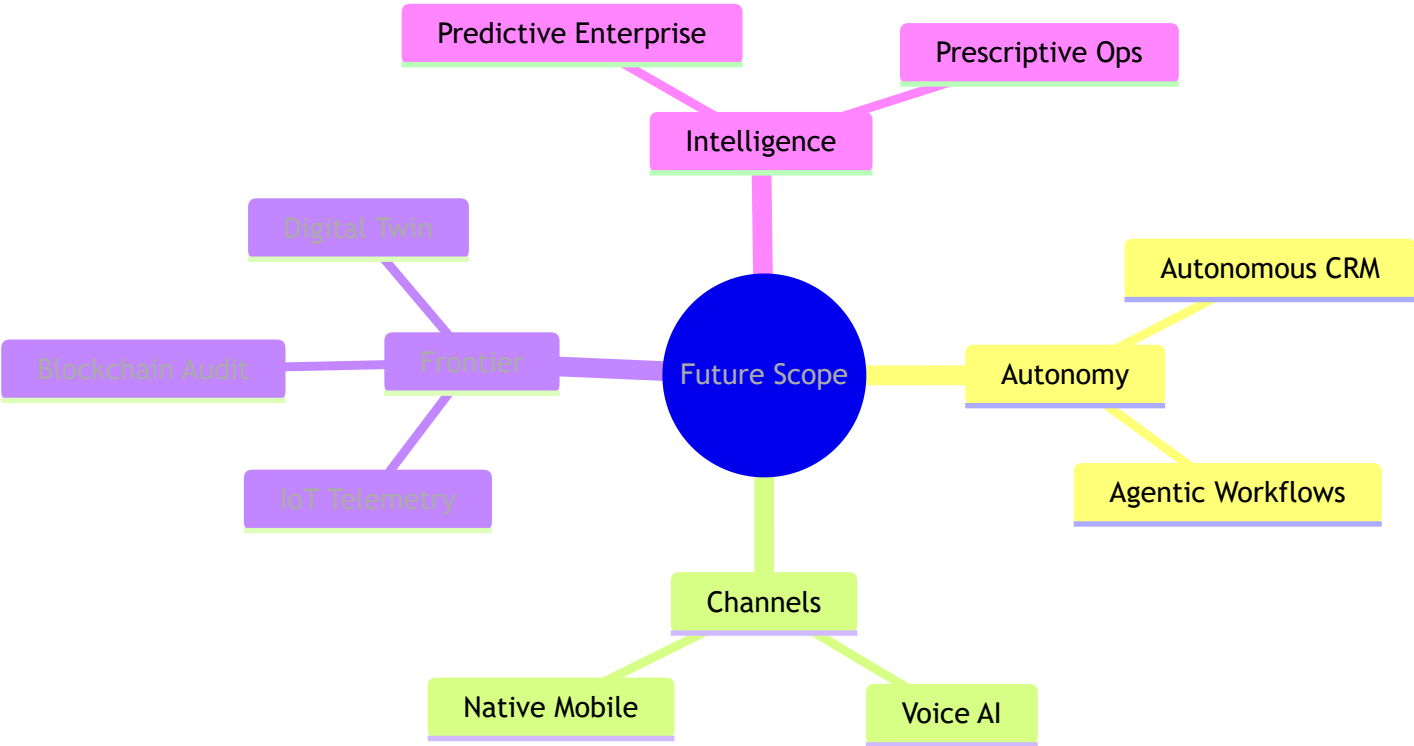
7.1 In Scope

The initial baseline release delivers the 8-Level operating system, nine BIOMs (Sales, Marketing, Finance, HRMS, Commerce, Service, Project, Vendor, Partner), the Workflow Engine, the AI Intelligence Mesh, the Executive Intelligence Center, multi-tenancy, RBAC, audit logging, universal search, the integration/API platform, and billing/tenant management.

7.2 Out of Scope (Baseline)

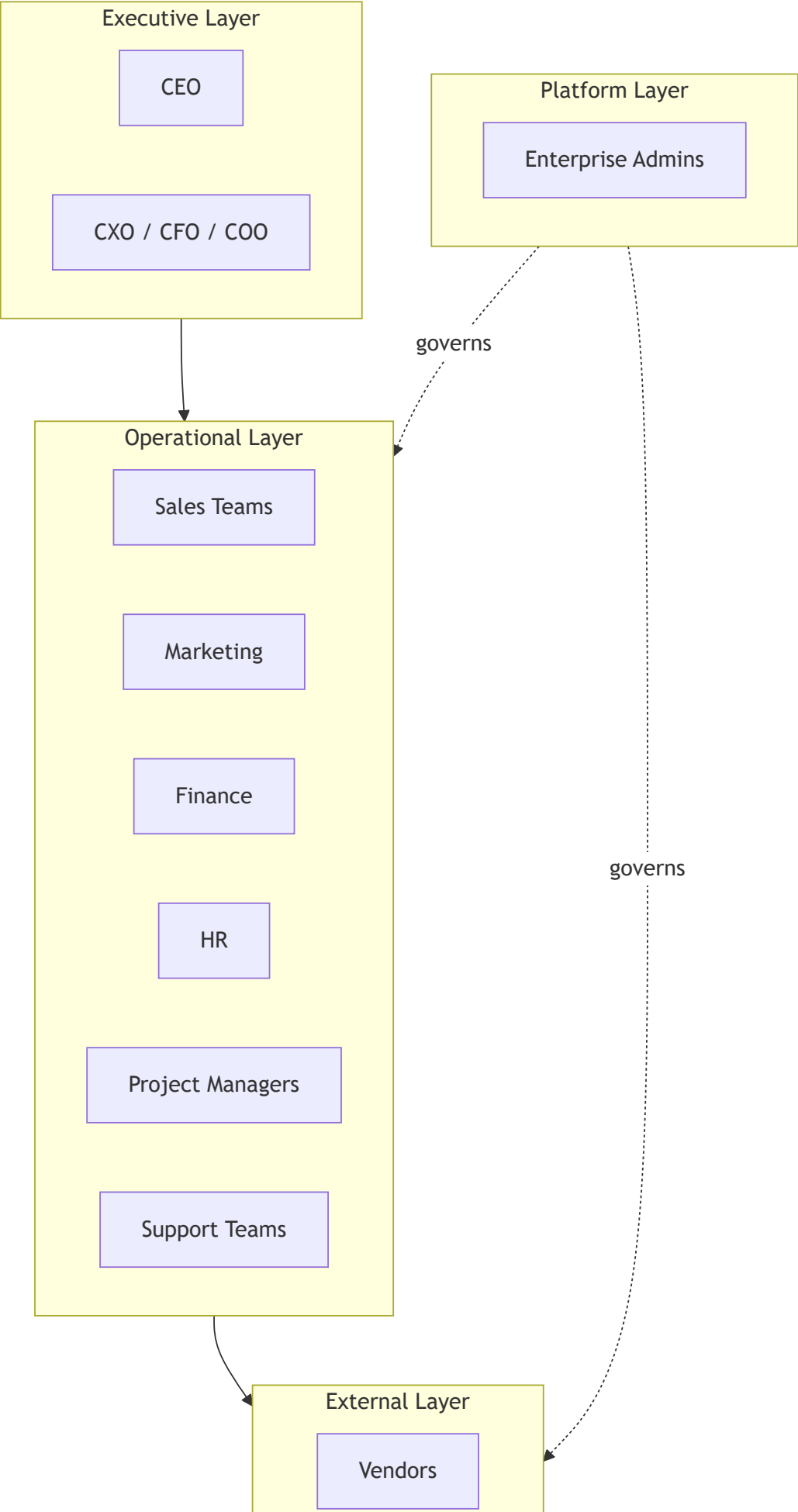
Bespoke industry verticalizations (e.g., regulated healthcare claims adjudication), on-device mobile-native apps (web-responsive only at baseline), hardware/IoT edge gateways, and blockchain settlement are explicitly deferred.

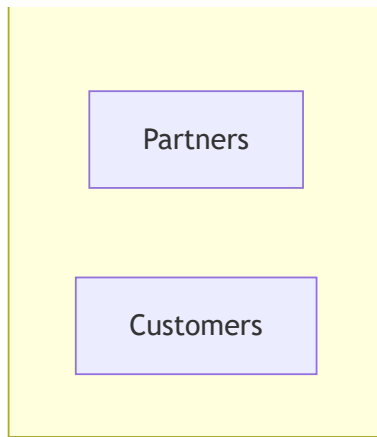
7.3 Future Scope



8. Target Users

8.1 Persona Map





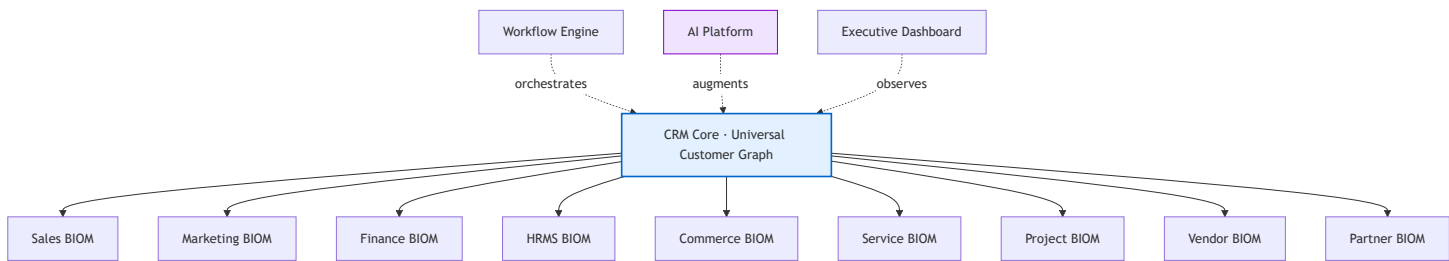
8.2 Persona Needs

Persona	Primary Need	Horizon 360 Answer
CEO	One truthful view of the business.	Executive Intelligence Center
CXO / CFO	Real-time financial & operational control.	Finance BIOM + live analytics
Sales	Pipeline clarity and less admin.	Sales BIOM + Copilot
Marketing	Attribution tied to revenue.	Marketing BIOM on shared graph
Finance	Accurate, auditable books.	Finance BIOM + audit logs
HR	Unified employee lifecycle.	HRMS BIOM
Project Managers	Delivery visibility + resourcing.	Project BIOM
Support	Full customer context per ticket.	Service BIOM + 360 view
Vendors / Partners	Self-service portals.	Vendor & Partner BIOMs
Customers	Coherent experience across touchpoints.	Unified customer node
Enterprise Admins	Governance, security, tenancy.	RBAC, audit, tenant mgmt

9. Product Features

Horizon 360 ships a comprehensive module estate. Each is a first-class citizen of the platform, not a plugin.

9.1 Module Overview



9.2 Feature Catalogue

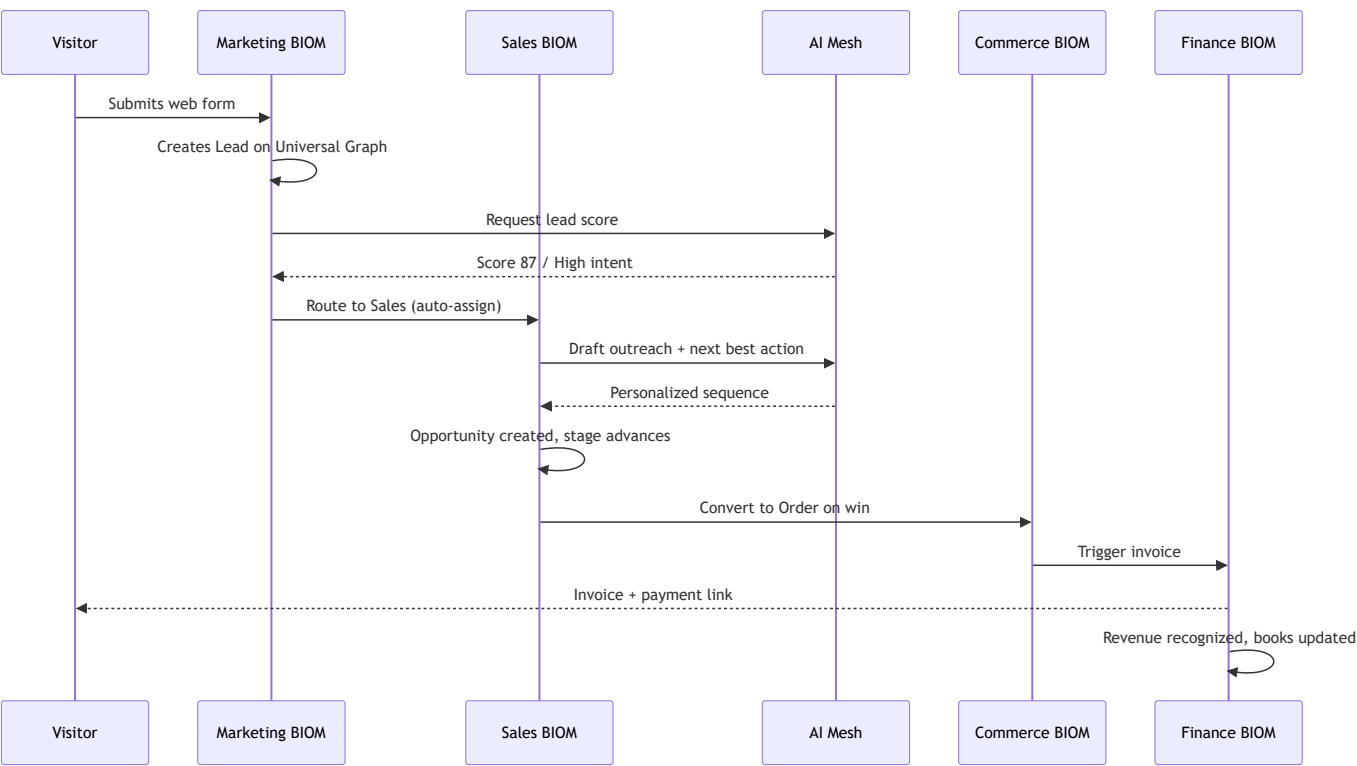
Module	Core Capabilities
CRM Core	Accounts, contacts, leads, opportunities, activity timeline, the universal customer graph.
Sales BIOM	Pipeline & stages, forecasting, quotes/CPQ, territory & quota, deal rooms, AI deal scoring.
Marketing BIOM	Campaigns, segmentation, journeys, email/SMS, lead scoring, attribution on the shared graph.
Finance BIOM	GL, AR/AP, invoicing, payments, revenue recognition, multi-currency, dunning, tax.
HRMS BIOM	Employee lifecycle, org chart, leave, payroll inputs, performance, onboarding/offboarding.
Commerce BIOM	Catalog, cart, orders, fulfillment, pricing, promotions, subscriptions.
Service BIOM	Omnichannel ticketing, SLAs, knowledge-grounded responses, CSAT, escalations.
Project BIOM	Projects, tasks, Gantt, resourcing, time tracking, billing integration.
Vendor BIOM	Vendor master, procurement, POs, vendor portal, performance scoring.
Partner BIOM	Partner onboarding, deal registration, commissions, partner portal.
Workflow Engine	Visual builder, durable executions, approvals, escalations, SLAs, human tasks.
AI Platform	Executive Copilot, per-BIOM agents, RAG, semantic search, forecasting, guardrails.
Executive Dashboard	Cross-BIOM KPIs, drill-downs, real-time tiles, anomaly alerts.
Analytics Center	Self-serve exploration on ClickHouse, cohorts, funnels, retention.
Knowledge Base	Internal/external articles, versioning, AI-grounded retrieval.

Module	Core Capabilities
Document Intelligence	OCR, extraction, classification, contract data capture.
Notification Center	Multi-channel, preference-aware, digest & real-time.
Universal Search	One search bar across every entity (Elasticsearch + semantic).
Role Management	RBAC + ABAC, fine-grained permissions, delegation.
Audit Logs	Immutable, queryable, per-tenant, compliance-grade.
Integrations	Connectors, webhooks, event subscriptions.
API Platform	REST + gRPC, versioned, rate-limited, documented.
Billing	Metered usage, seat & module billing, AI credits.
Tenant Management	Provisioning, isolation, lifecycle, configuration.

📌 **Important Highlight:** Every module above writes to the *same* Universal Data Model. There is no "sync" between Sales and Finance — they read the same opportunity and the same invoice.

10. Detailed User Journey

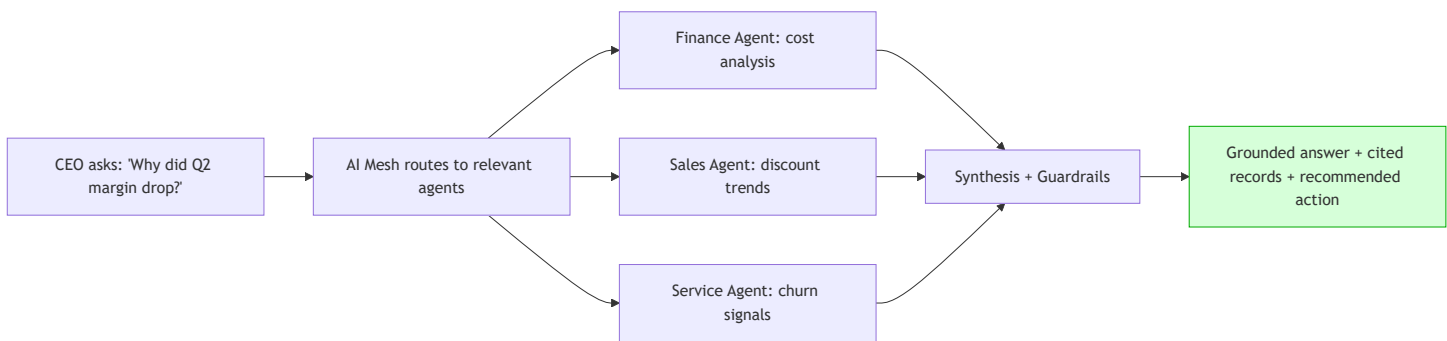
10.1 Scenario: Lead to Cash (End-to-End)



10.2 Journey Library

Journey	Trigger	Outcome
Lead Generation	Form / ad / event	Scored lead on the graph, routed instantly.
Sales Pipeline	Lead conversion	Forecasted opportunity with AI guidance.
Customer Support	Inbound ticket	Full-context resolution, KB-grounded reply.
Hiring	Job requisition	Candidate pipeline → onboarding workflow.
Project Management	Won deal / SOW	Project spun up, resourced, tracked, billed.
Procurement	Purchase request	Vendor selection → PO → approval → receipt.
Billing	Order / milestone	Invoice → payment → reconciliation.
AI Copilot	Natural-language ask	Grounded answer + executable action.
Executive Dashboard	Login	Real-time cross-business state.

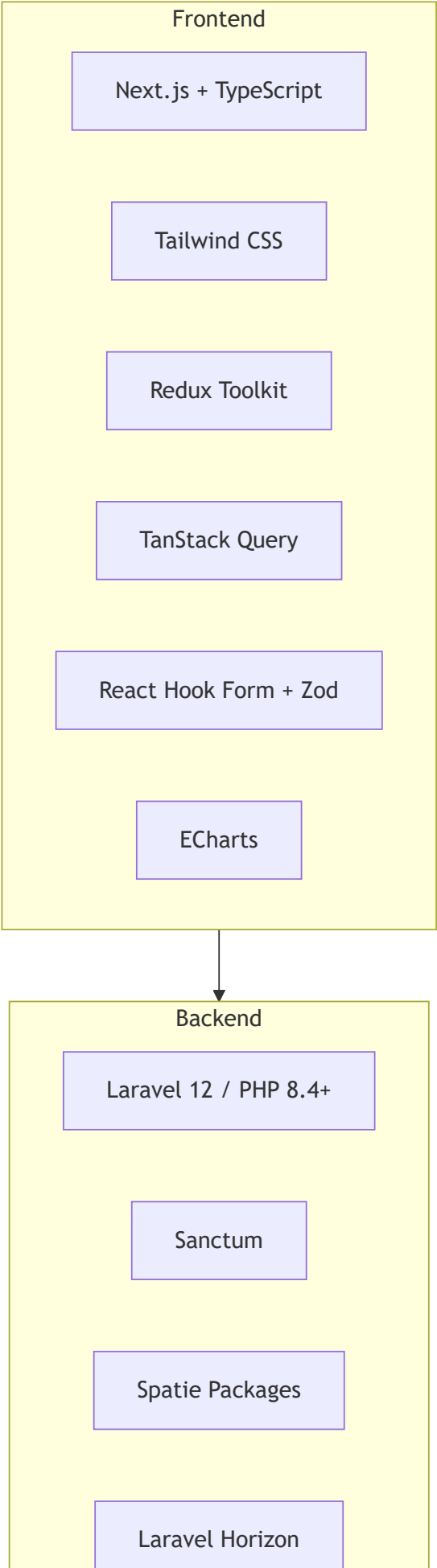
10.3 Executive Copilot Journey

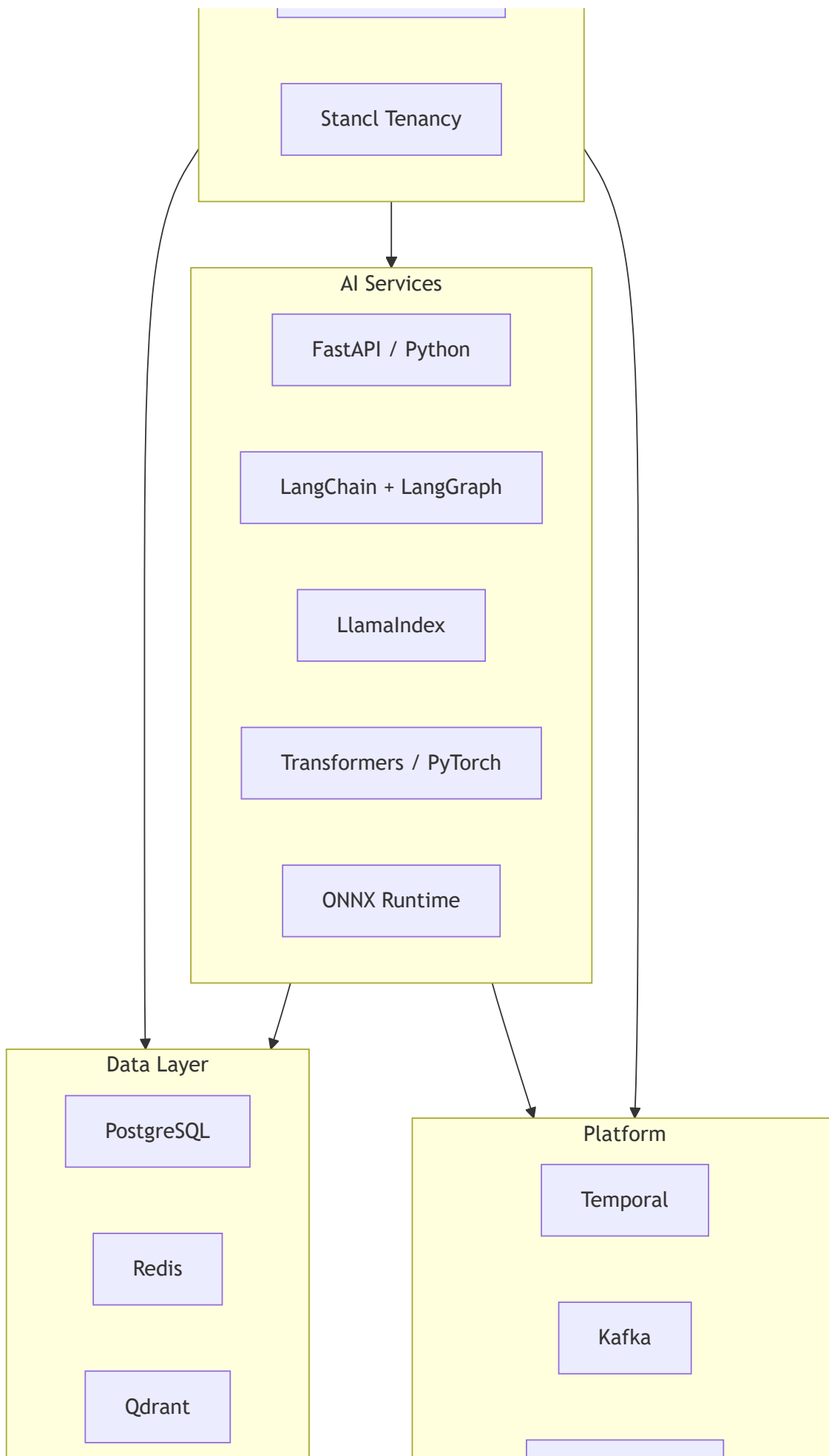


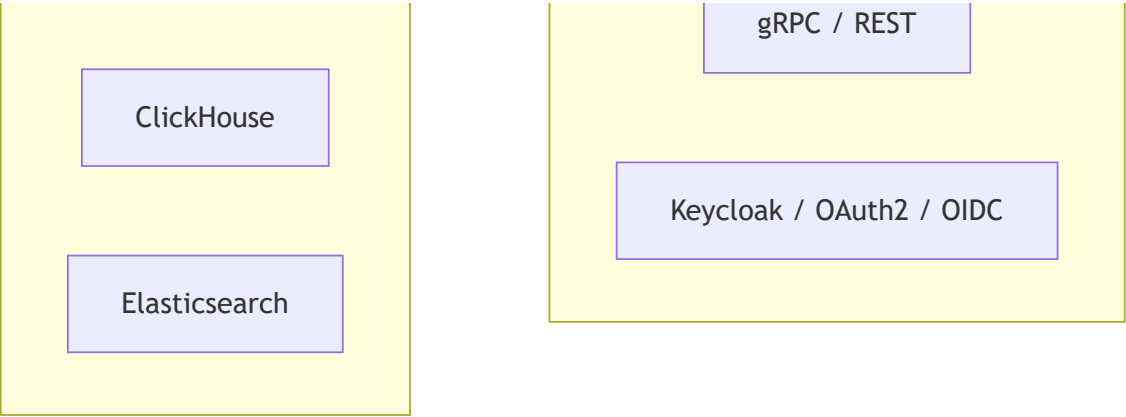
11. Complete Technology Stack

The stack below is the **official Horizon 360 stack**. Each selection is justified by a concrete architectural requirement.

11.1 Stack Map







11.2 Frontend — and Why

Technology	Why Selected
Next.js	SSR/ISR for fast first paint on data-heavy enterprise dashboards; file-based routing; edge-ready.
TypeScript	Type safety across a very large surface area reduces runtime defects in a mission-critical app.
Tailwind CSS	Utility-first system enforces a consistent design language at enterprise scale without CSS sprawl.
Redux Toolkit	Predictable, debuggable client state for complex multi-module UIs.
TanStack Query	Server-state caching, background refetch, and stale-while-revalidate for live data.
React Hook Form + Zod	Performant forms with schema validation shared between client and contract.
ECharts	High-performance, richly interactive charts for the Analytics Center and dashboards.

11.3 Backend — and Why

Technology	Why Selected
Laravel 12 / PHP 8.4+	Mature, batteries-included framework; rapid domain modeling; strong ecosystem for multi-tenant SaaS.
Sanctum	Lightweight token/SPA auth for first-party clients.
Spatie Packages	Battle-tested permissions, activity logs, media, and query builders.
Laravel Horizon	Visibility and control over Redis-backed queues at scale.

Technology	Why Selected
StancI Tenancy	Robust multi-tenancy (database/schema isolation) — the backbone of the SaaS model.

11.4 AI Services — and Why

Technology	Why Selected
FastAPI / Python	High-throughput async services; the native habitat of the ML ecosystem.
LangChain + LangGraph	Composable chains and stateful, graph-based agent orchestration for multi-step reasoning.
LlamaIndex	Best-in-class data framework for RAG ingestion, indexing, and retrieval.
Transformers / PyTorch	Custom model fine-tuning, embeddings, and inference.
ONNX Runtime	Optimized, portable inference for latency-sensitive in-house models.

11.5 Databases — and Why

Technology	Role	Why Selected
PostgreSQL	OLTP system of record	ACID, relational integrity, JSONB flexibility, extensions.
Redis	Cache + queues	Sub-millisecond cache, queue backend, rate limiting.
Qdrant	Vector store	Semantic search and AI memory at scale.
ClickHouse	OLAP analytics	Columnar speed for real-time analytics over billions of rows.
Elasticsearch	Search	Full-text universal search across all entities.

11.6 Platform, Security, DevOps, Testing — and Why

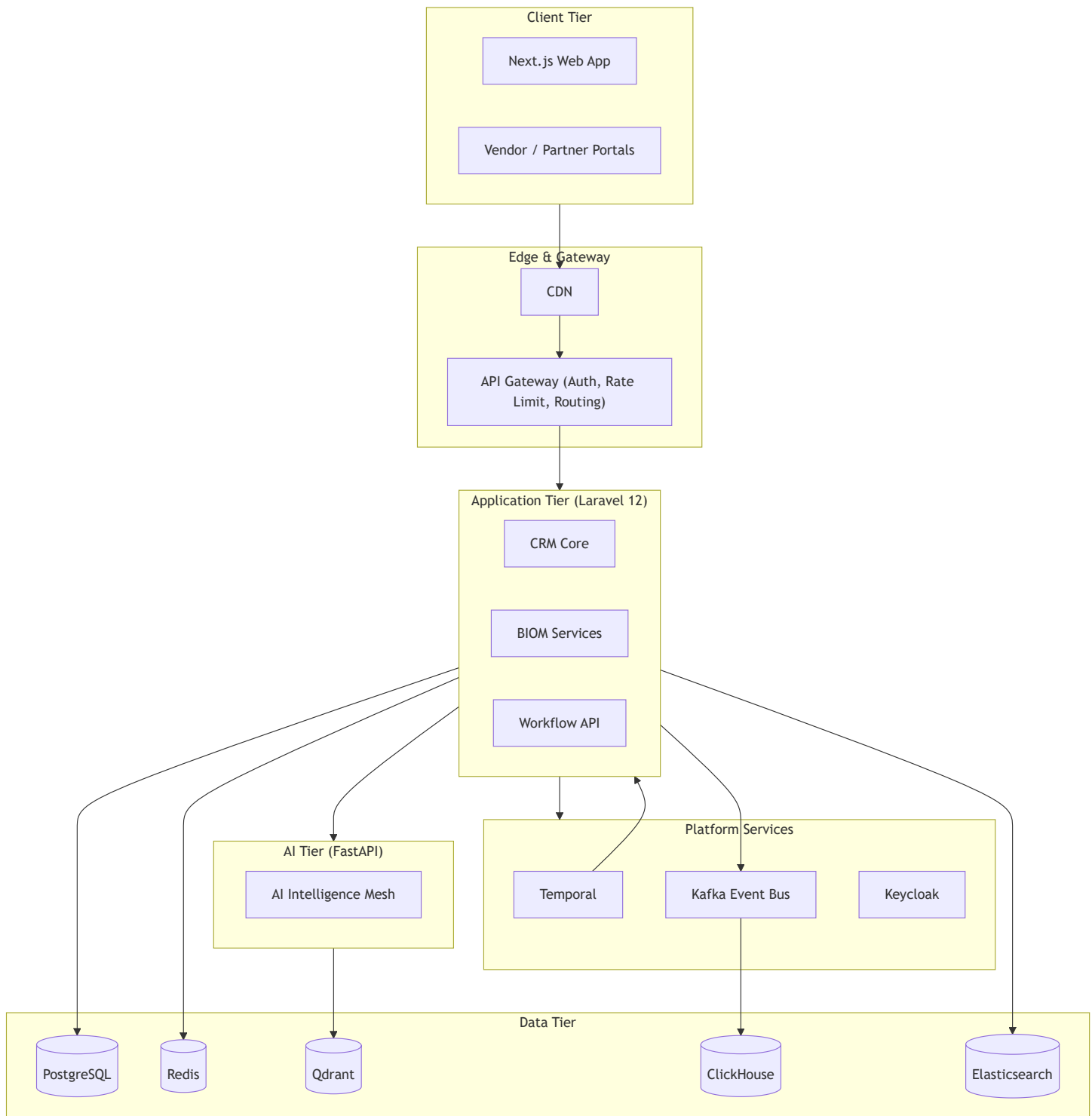
Layer	Technology	Why Selected
Workflow	Temporal	Durable, fault-tolerant orchestration of long-running business processes.
Communication	Kafka, gRPC, REST	Event backbone (Kafka), low-latency internal RPC (gRPC), open external contracts (REST).

Layer	Technology	Why Selected
Security	Keycloak, OAuth2, OIDC, MFA	Standards-based identity, SSO, and strong auth.
Deployment	Docker, Kubernetes, Terraform, Vault	Containerized, orchestrated, declaratively provisioned, with managed secrets.
Monitoring	ELK, Prometheus, Grafana, OpenTelemetry	Logs, metrics, dashboards, and distributed tracing.
Testing	Playwright, PHPUnit, Pytest, k6	E2E, backend unit, AI/Python tests, and load testing.

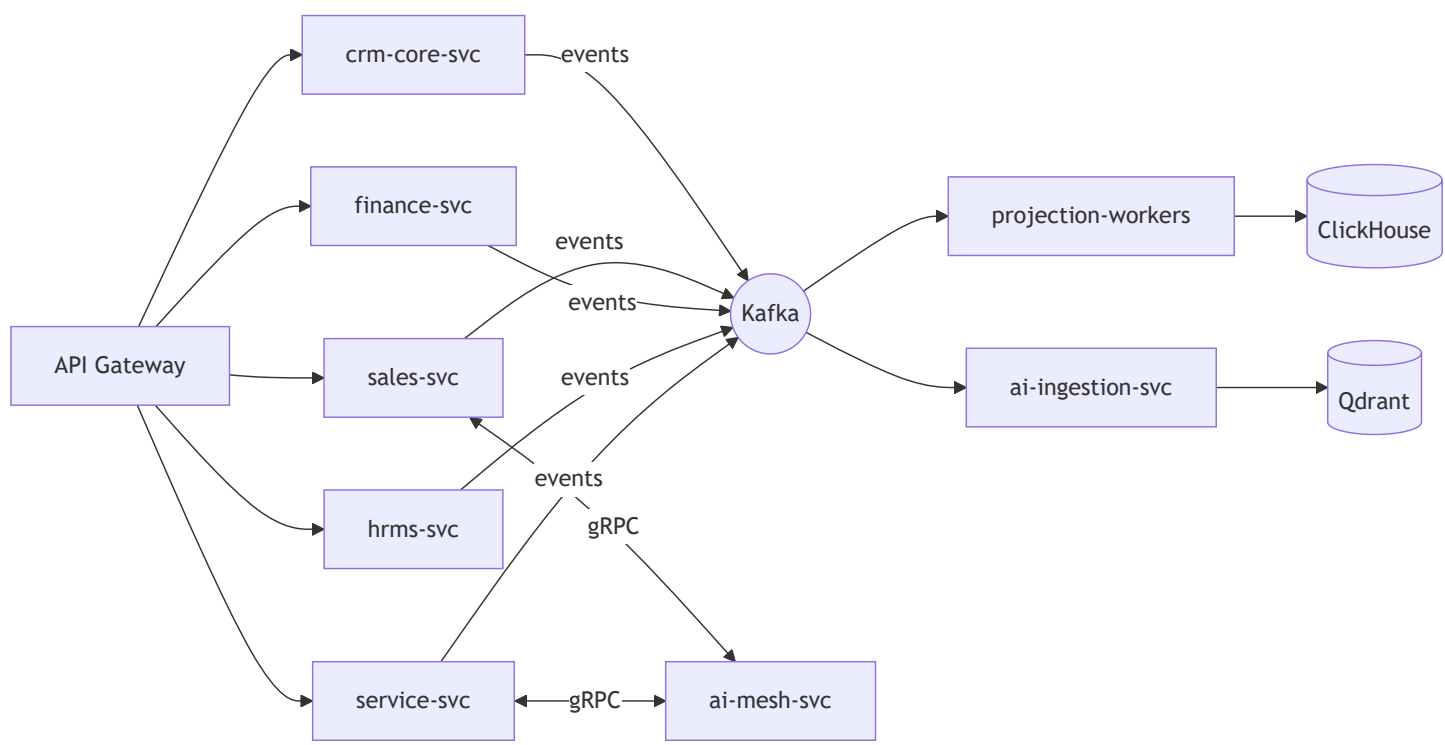
 **Best Practice Note:** The stack deliberately separates the OLTP truth (PostgreSQL) from analytics (ClickHouse), search (Elasticsearch), and vectors (Qdrant). Polyglot persistence is intentional — each engine does what it is best at, fed from one event stream.

12. System Architecture

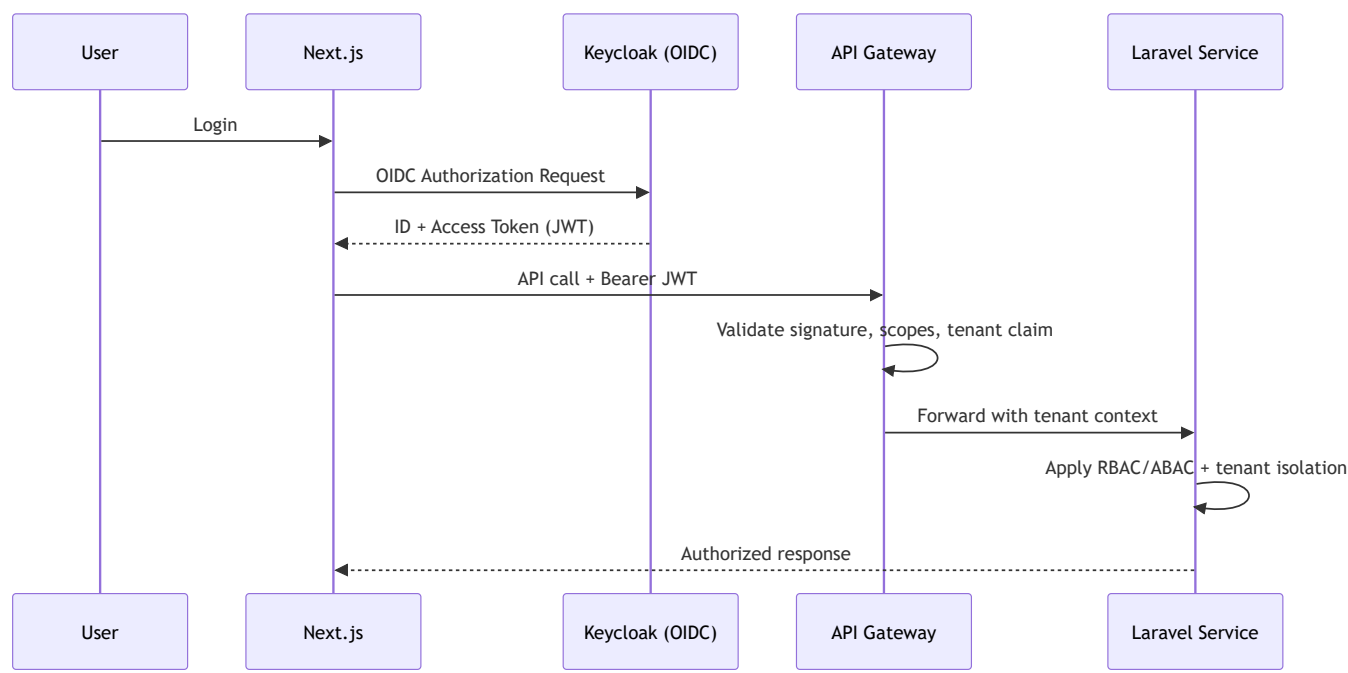
12.1 High-Level Architecture



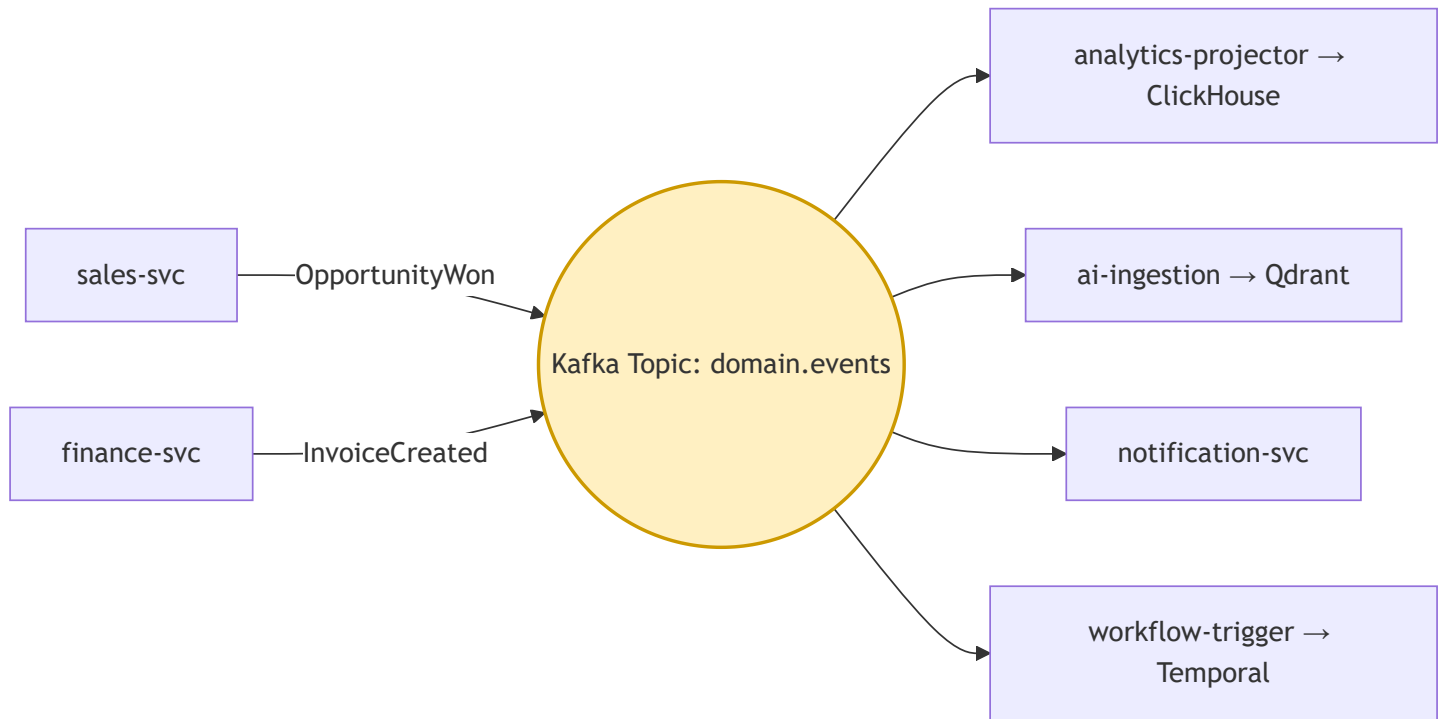
12.2 Low-Level / Microservice Architecture



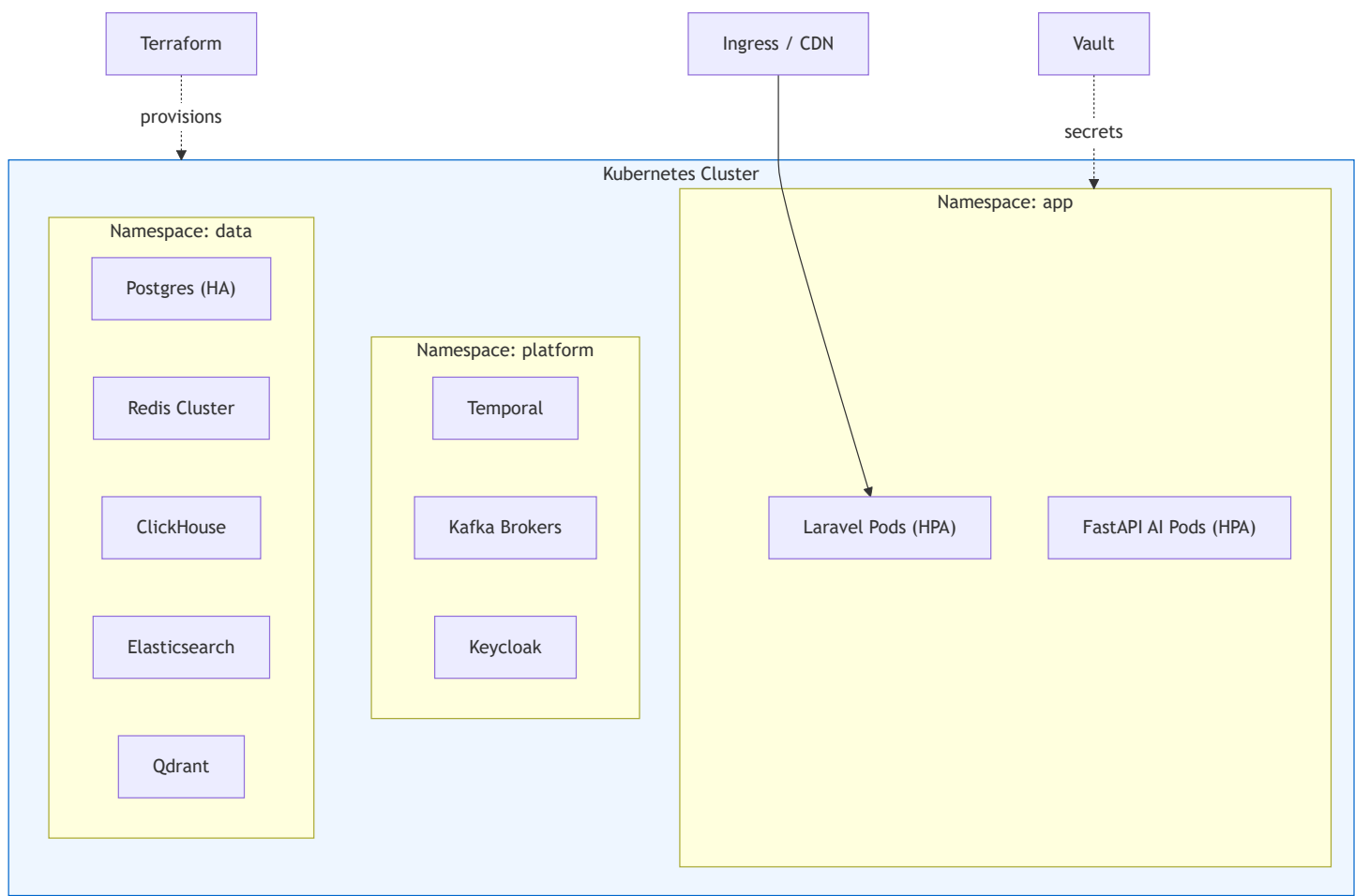
12.3 Authentication Flow



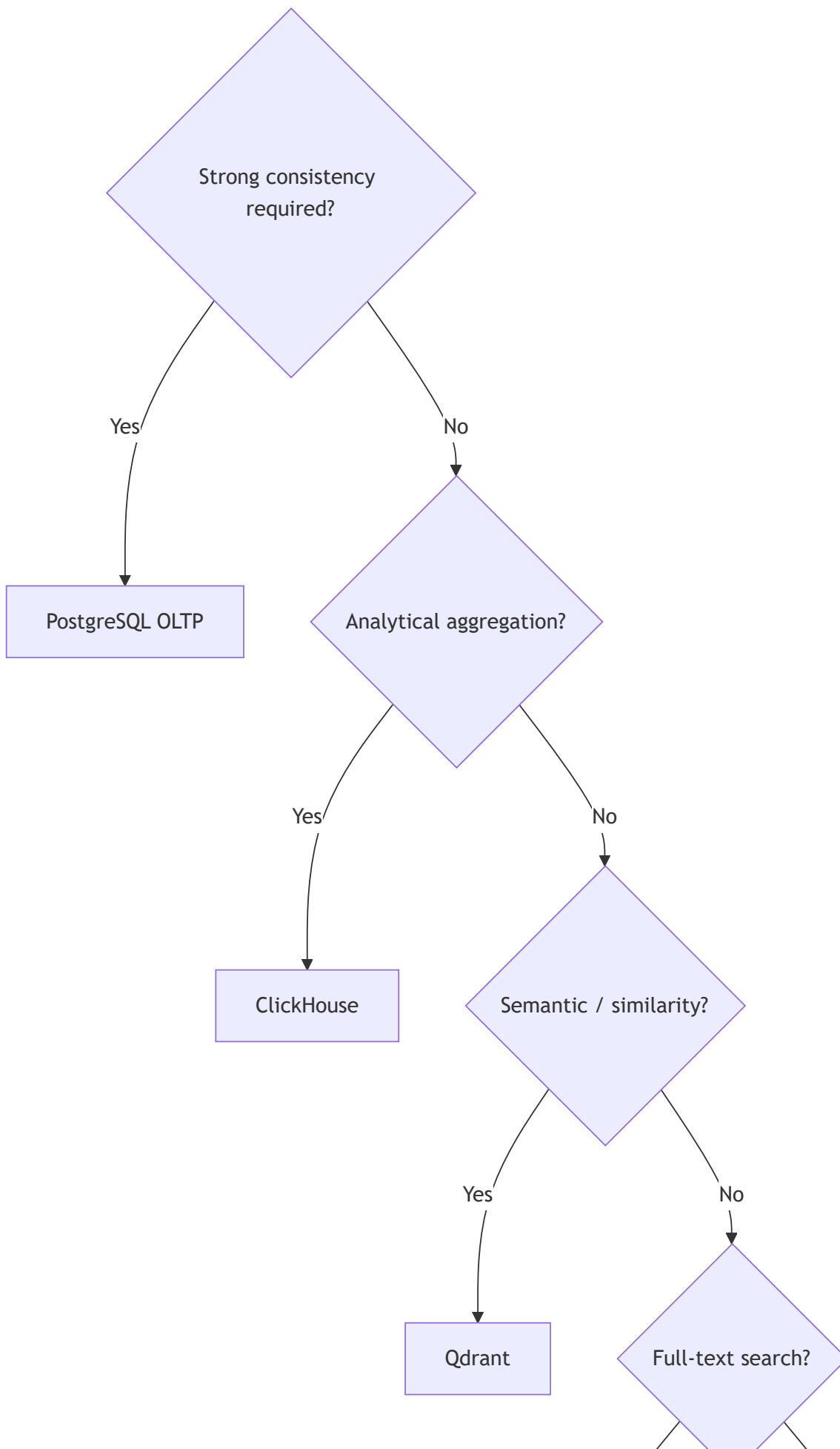
12.4 Event-Driven Architecture

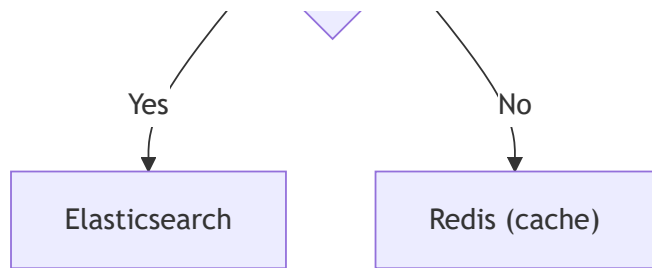


12.5 Deployment Architecture



12.6 Architectural Decision Tree

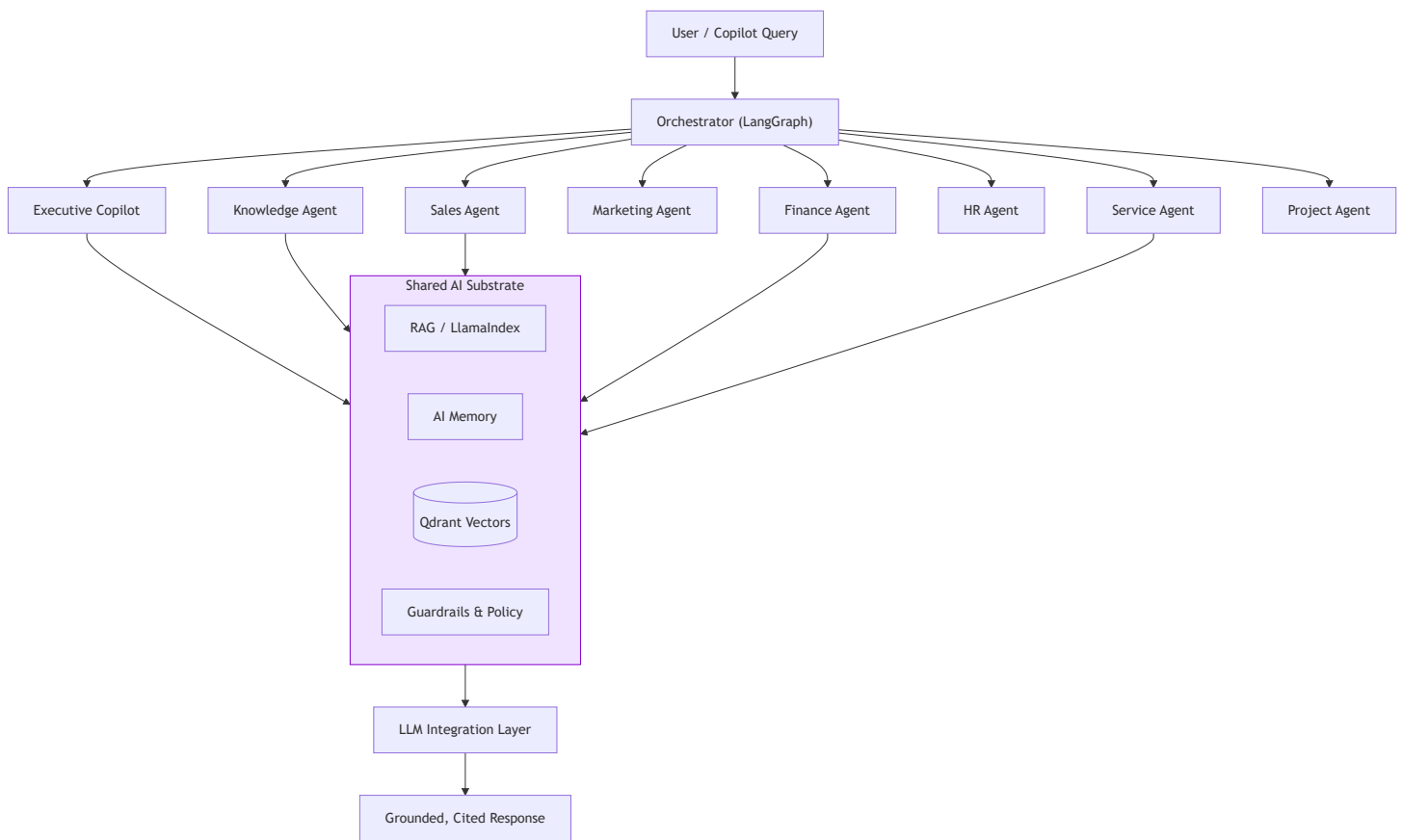




13. AI Architecture

13.1 The AI Intelligence Mesh

The Mesh is a federation of domain agents that share a common memory, retrieval layer, and guardrail policy. An orchestrator (LangGraph) routes a request to the right agent(s) and synthesizes a grounded answer.

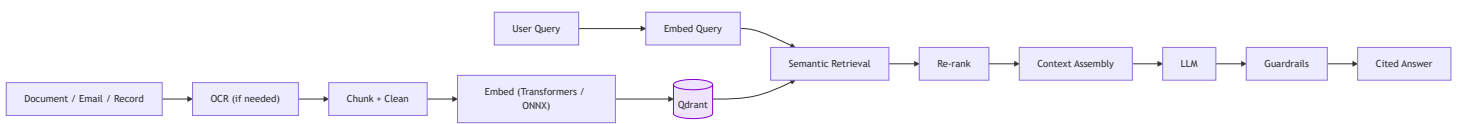


13.2 Agent Responsibilities

Agent	Responsibility
Executive Copilot	Cross-BIOM synthesis; answers strategic questions with cited evidence.

Agent	Responsibility
Knowledge Agent	Retrieves and grounds answers in the Knowledge Base.
Sales Agent	Deal scoring, next-best-action, outreach drafting, forecast explanation.
Marketing Agent	Segmentation, content drafting, journey suggestions, attribution narration.
Finance Agent	Anomaly detection, variance analysis, dunning suggestions.
HR Agent	Policy Q&A, candidate screening assistance, onboarding orchestration.
Service Agent	KB-grounded responses, ticket triage, sentiment & escalation.
Project Agent	Risk flags, resourcing suggestions, status summarization.

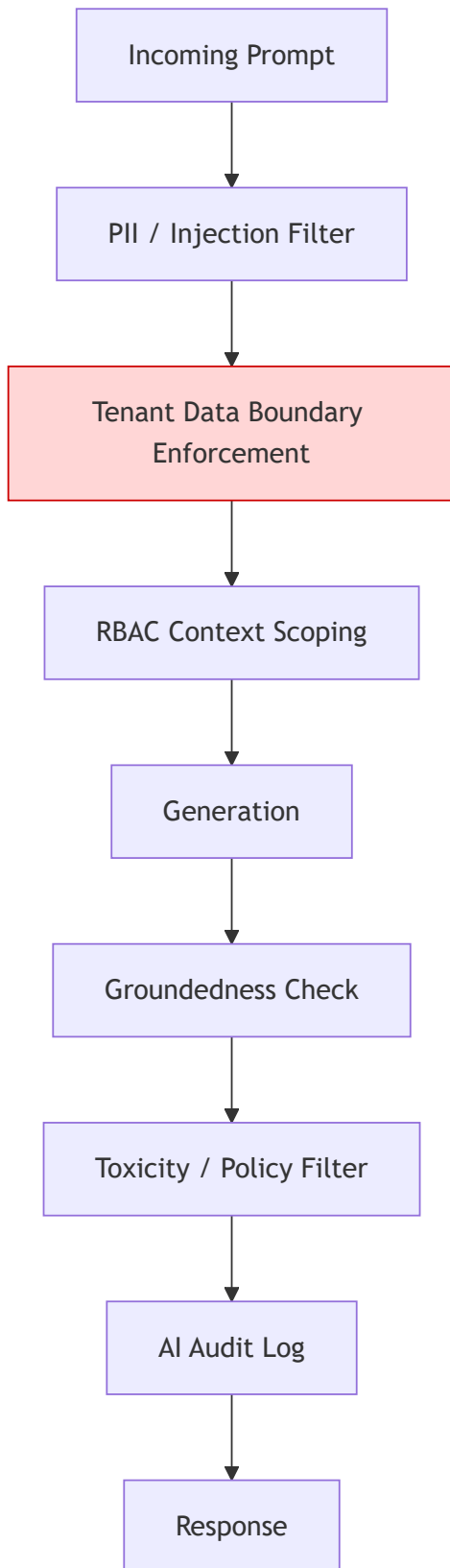
13.3 RAG & Document Intelligence Pipeline



13.4 Forecasting & Recommendation

The Mesh hosts time-series forecasting (pipeline, revenue, churn) and a recommendation engine (next-best-action, cross-sell). Models run on PyTorch for training and ONNX Runtime for low-latency inference, with predictions written back to the Universal Data Model as first-class, explainable signals.

13.5 Guardrails & AI Security

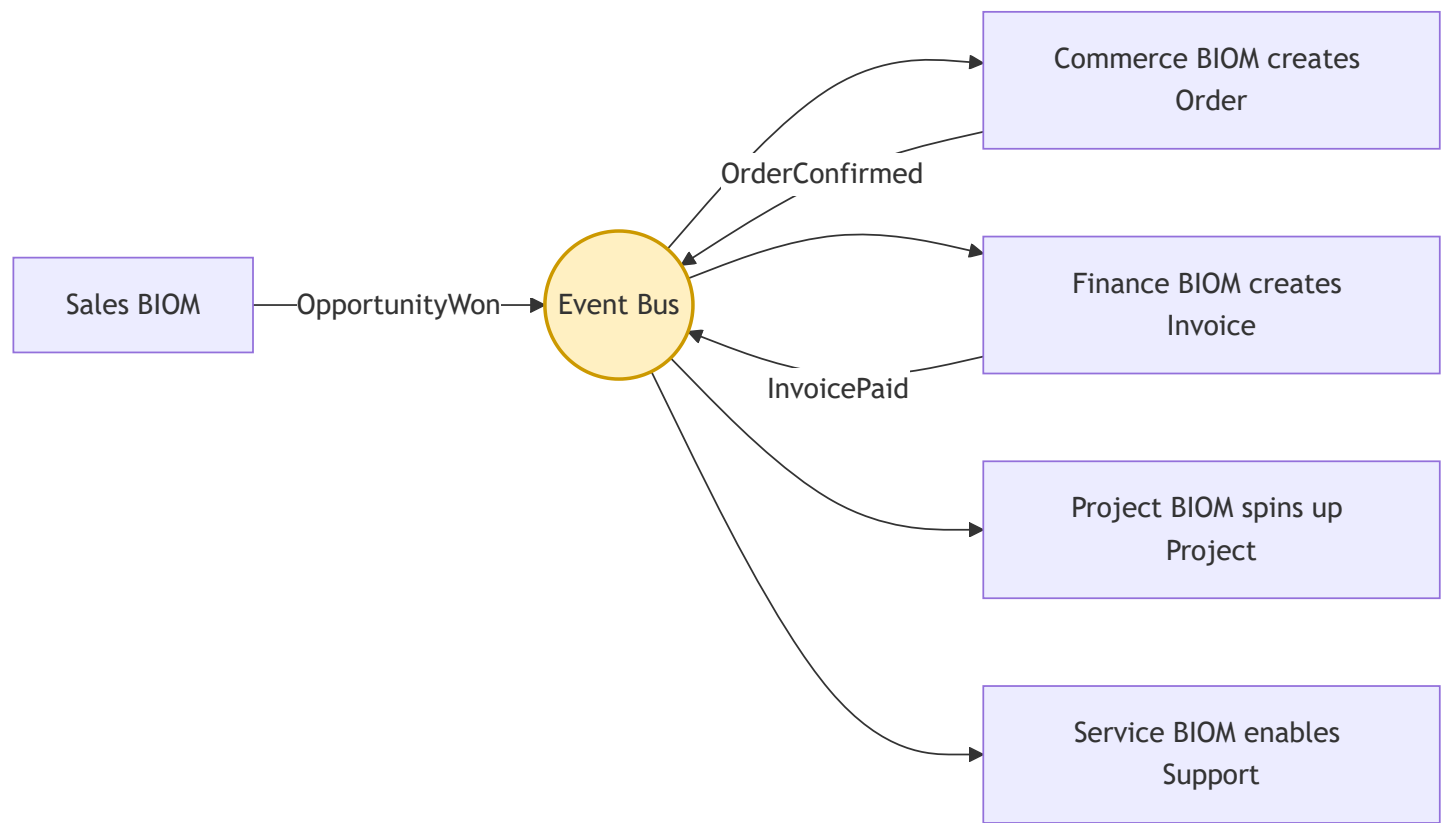


⚠ Important Highlight: Every AI call is scoped to the requesting user's RBAC permissions and tenant boundary *before* retrieval. The AI can never surface a record the user is not authorized to see.

14. BIOM Ecosystem

A **BIOM (Business Intelligent Operating Module)** is a self-contained domain that shares the platform substrate. Each BIOM owns its bounded context, publishes events, exposes APIs, and is augmented by its Mesh agent.

14.1 Cross-BIOM Communication

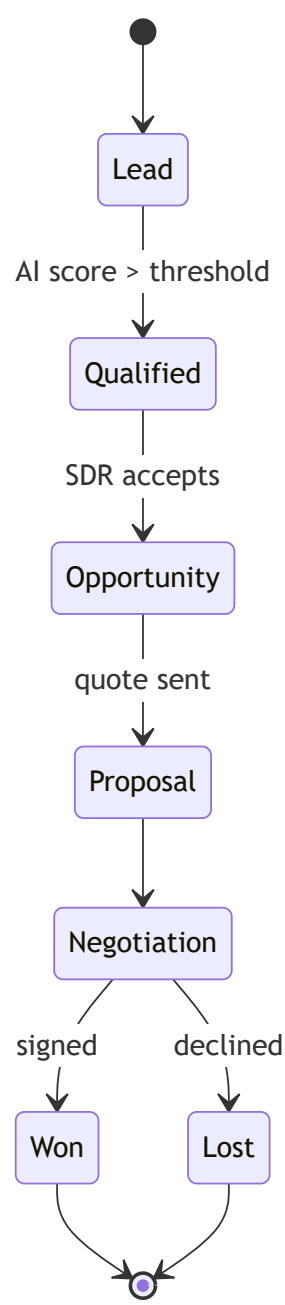


14.2 BIOM Specifications

BIOM	Responsibilities	Key Entities	Publishes
Sales	Pipeline, forecast, CPQ.	Lead, Opportunity, Quote	OpportunityWon, QuoteSent
Marketing	Campaigns, journeys, scoring.	Campaign, Segment, Journey	LeadScored, CampaignSent
Finance	GL, AR/AP, invoicing.	Invoice, Payment, Account	InvoiceCreated, InvoicePaid
HRMS	Employee lifecycle.	Employee, Position, Leave	EmployeeHired, LeaveApproved
Commerce	Catalog, orders.	Product, Order, Cart	OrderConfirmed, OrderShipped

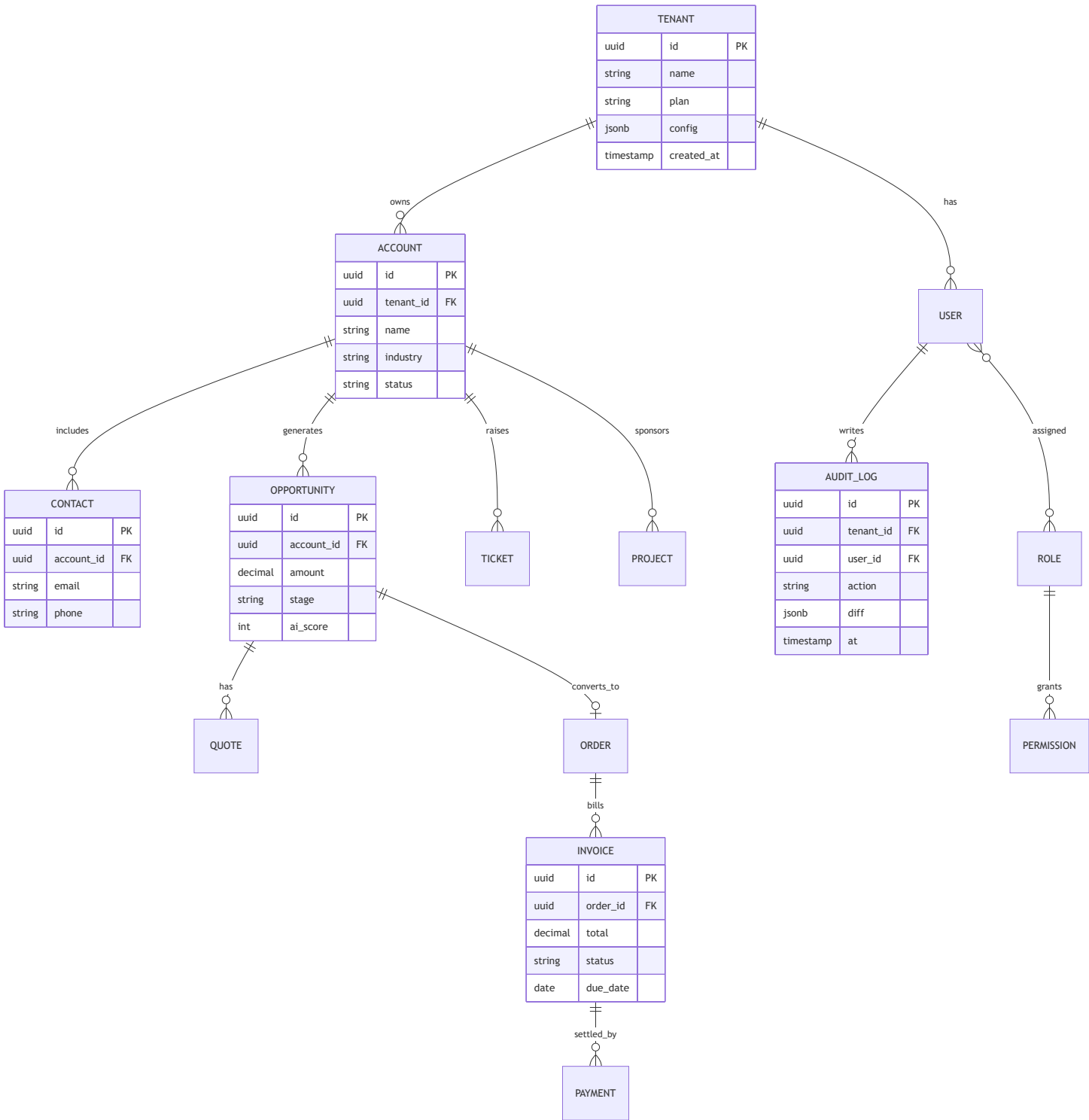
BIOM	Responsibilities	Key Entities	Publishes
Service	Ticketing, SLAs.	Ticket, SLA, Article	TicketOpened, TicketResolved
Project	Delivery, resourcing.	Project, Task, Timesheet	ProjectCreated, TaskCompleted
Vendor	Procurement.	Vendor, PO, Receipt	POIssued, GoodsReceived
Partner	Channel mgmt.	Partner, DealReg, Commission	DealRegistered, CommissionPaid

14.3 Single BIOM Internal Workflow (Sales Example)



15. Database Design

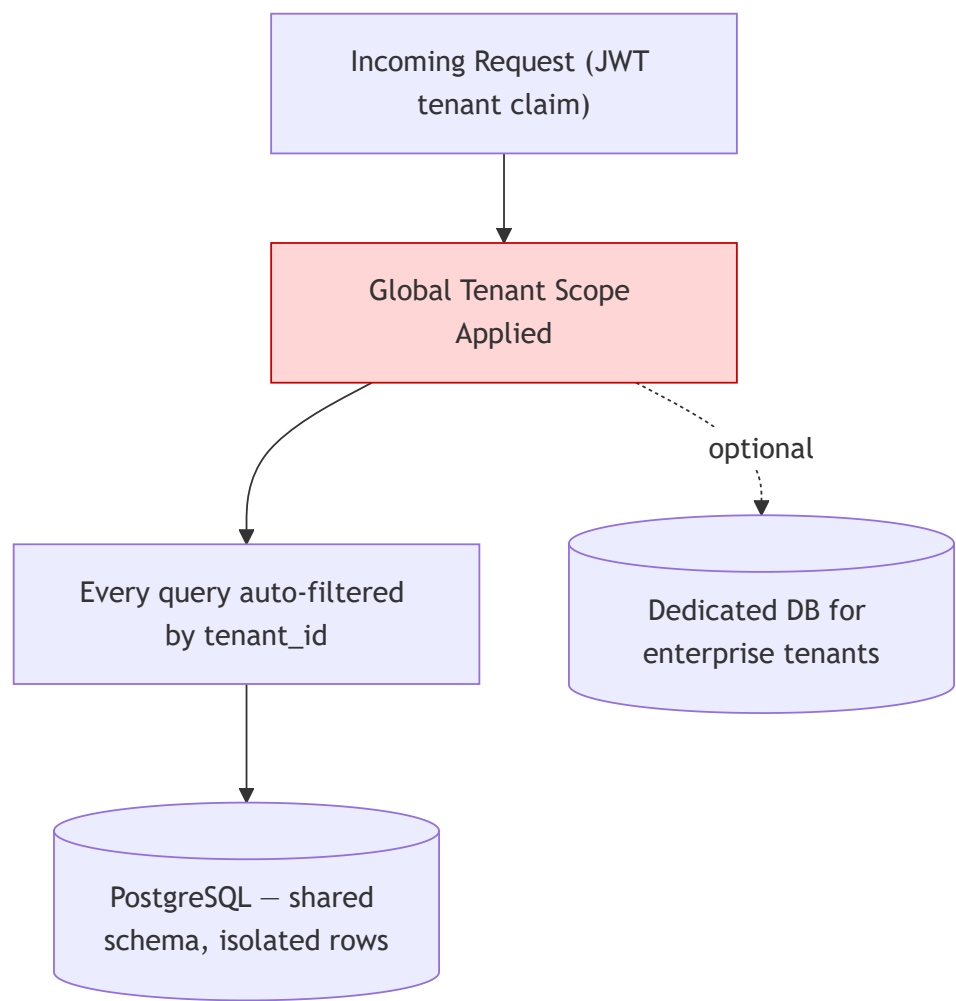
15.1 Universal Data Model — Core ER Diagram



15.2 Keys, Indexes & Integrity

Concern	Approach
Primary Keys	UUID v7 (time-sortable) for global uniqueness and shard-friendliness.
Foreign Keys	Enforced at DB level for OLTP integrity; cascade rules per relationship.
Indexes	B-tree on FK + status columns; GIN on JSONB config; partial indexes on hot statuses.
Tenant Isolation	<code>tenant_id</code> on every row (Stancl), enforced via global query scopes + row-level checks.
Audit Logs	Append-only table; immutable; partitioned monthly; queried separately.
Permissions	Role/permission tables (Spatie), evaluated per request with ABAC overlays.

15.3 Tenant Isolation Model



💡 Best Practice Note: Horizon 360 supports a hybrid tenancy model — shared-schema row isolation for the long tail, and dedicated-database isolation for large enterprise tenants with regulatory requirements. The application code is identical; only the connection resolver differs.

16. Functional Requirements

Functional requirements are stated module-wise using the convention **FR-[MODULE]-[n]**. Each is testable and traceable.

16.1 CRM Core

ID	Requirement
FR-CRM-1	The system shall maintain exactly one canonical record per real-world entity (account, contact).
FR-CRM-2	The system shall render a unified activity timeline aggregating events from all BIOMs.
FR-CRM-3	The system shall support merge/deduplication with audit trail.

16.2 Sales BIOM

ID	Requirement
FR-SAL-1	The system shall allow configurable pipeline stages per tenant.
FR-SAL-2	The system shall compute and display an AI deal score on every opportunity.
FR-SAL-3	The system shall generate quotes and convert won opportunities into orders automatically.

16.3 Finance BIOM

ID	Requirement
FR-FIN-1	The system shall generate invoices from orders and milestones.
FR-FIN-2	The system shall support multi-currency with daily FX rates.
FR-FIN-3	The system shall post double-entry GL transactions for every financial event.

16.4 Service BIOM

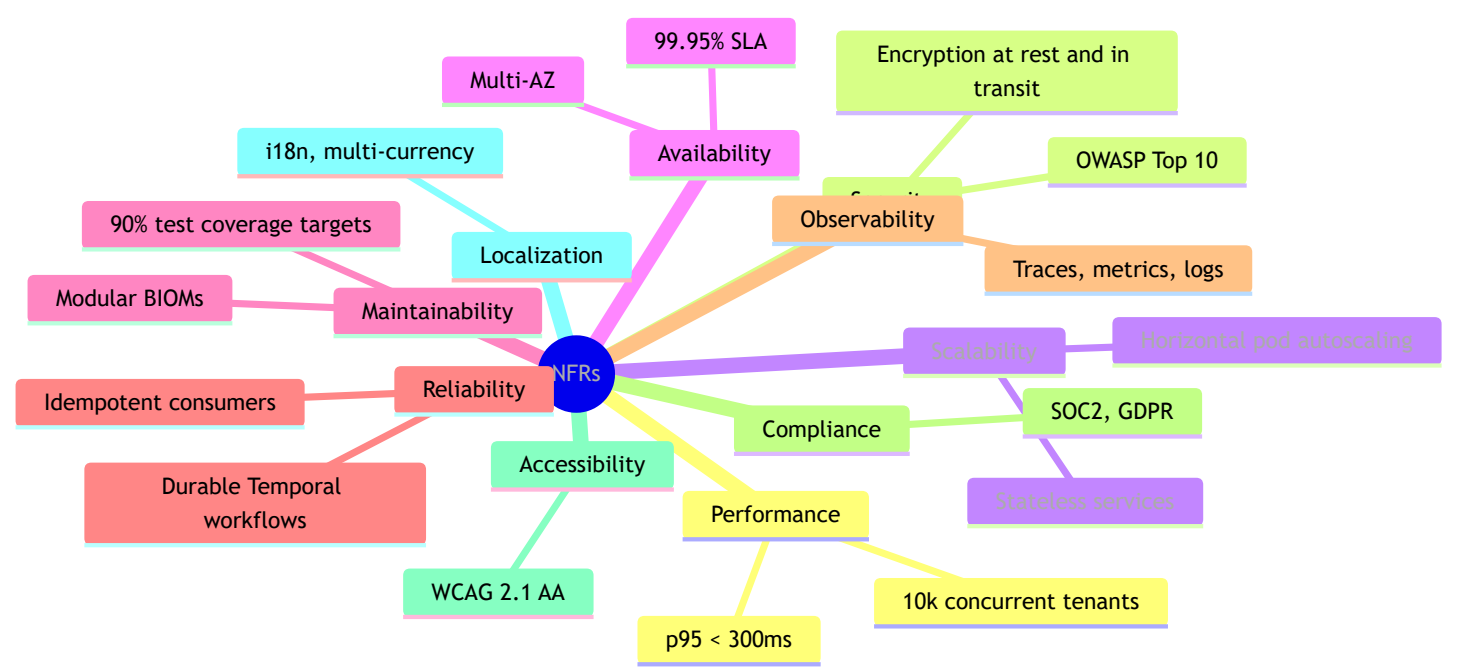
ID	Requirement
FR-SVC-1	The system shall capture tickets from email, web, and chat into one queue.
FR-SVC-2	The system shall enforce SLA timers with escalation on breach.

ID	Requirement
FR-SVC-3	The system shall propose KB-grounded responses via the Service Agent.

16.5 Platform-Wide

ID	Requirement
FR-PLT-1	The system shall provide universal search across all entities.
FR-PLT-2	The system shall record an immutable audit log for every mutating action.
FR-PLT-3	The system shall enforce RBAC + ABAC on every API call and AI retrieval.
FR-PLT-4	The system shall expose every capability via versioned REST/gRPC APIs.

17. Non-Functional Requirements



Attribute	Target	Mechanism
Performance	p95 < 300ms API; dashboards < 2s	Caching (Redis), ClickHouse, query tuning
Security	OWASP-clean; zero critical CVEs	SAST/DAST, dependency scanning, pen tests
Scalability	Linear to 10k+ tenants	Stateless services + K8s HPA
Availability	99.95%	Multi-AZ, health checks, graceful degradation

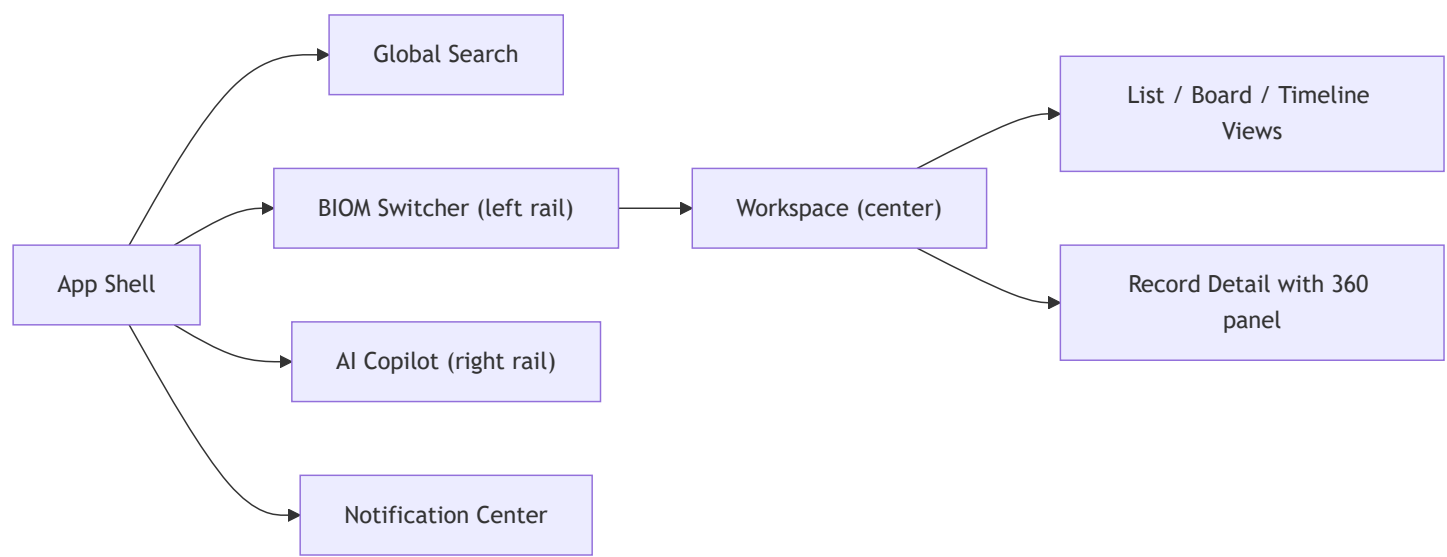
Attribute	Target	Mechanism
Maintainability	< 1 day onboarding per BIOM	Strict module boundaries, docs
Reliability	No lost business processes	Temporal durable execution
Observability	Full request tracing	OpenTelemetry, Grafana, ELK
Compliance	SOC2 Type II, GDPR	Audit logs, data residency, DPA
Accessibility	WCAG 2.1 AA	Semantic markup, contrast, keyboard nav
Localization	10+ locales	i18n framework, locale-aware formatting

18. UI/UX Planning

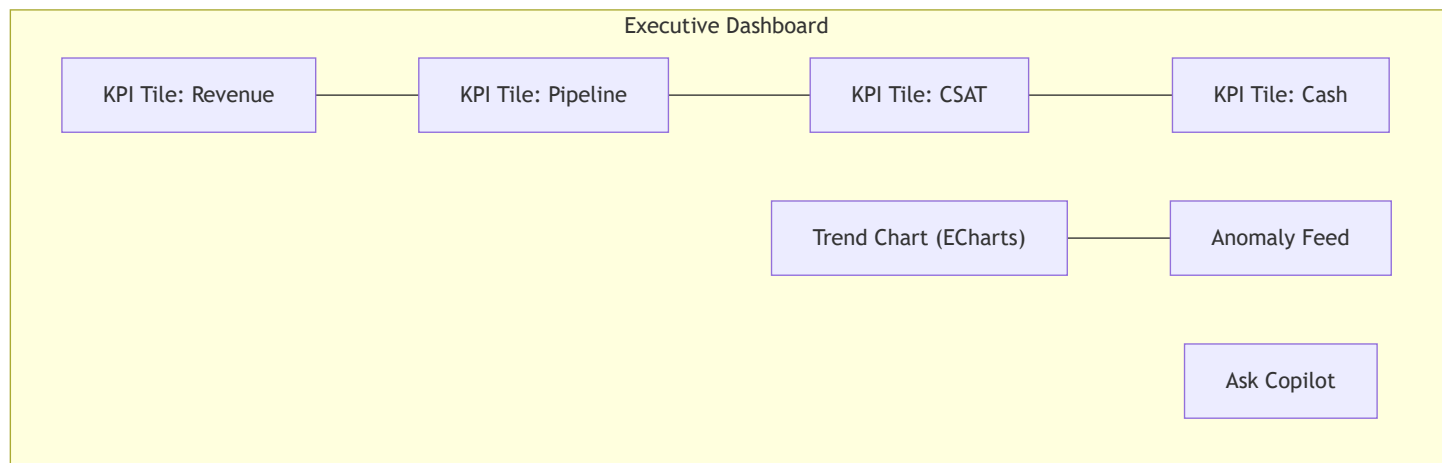
18.1 Design Philosophy

Horizon 360's interface follows three principles: **Clarity over density** (executives must grasp state in seconds), **Consistency across BIOMs** (one design system, learned once), and **Progressive disclosure** (power available, never overwhelming).

18.2 Navigation Model



18.3 Dashboard Layout



18.4 Design System & Responsiveness

The design system is built on Tailwind tokens (color, spacing, typography, elevation) consumed by a shared component library. All layouts are responsive down to tablet; critical read views (dashboards, approvals, notifications) are usable on mobile web. Accessibility is first-class: every interactive element is keyboard-navigable and screen-reader labeled, targeting WCAG 2.1 AA.

✦ **Important Highlight:** The AI Copilot is a persistent surface, not a separate page. Users invoke whole-business intelligence from inside any record they are working on.

19. API Design

19.1 Conventions

Concern	Convention
Style	Resource-oriented REST; gRPC for internal service-to-service.
Versioning	URI versioning: /api/v1/... ; deprecation headers on sunset.
Auth	OAuth2 / OIDC bearer tokens (Keycloak); Sanctum for first-party SPA.
Pagination	Cursor-based for large collections.
Errors	RFC 7807 Problem Details JSON.
Idempotency	Idempotency-Key header on POST for safe retries.

Concern	Convention
Rate Limiting	Per-tenant + per-key token buckets at the gateway.

19.2 Sample Endpoints

Method	Endpoint	Purpose
GET	/api/v1/accounts	List accounts (paginated, filterable).
POST	/api/v1/opportunities	Create an opportunity.
GET	/api/v1/opportunities/{id}	Retrieve one opportunity with AI score.
POST	/api/v1/invoices	Create an invoice from an order.
POST	/api/v1/ai/copilot/query	Submit a Copilot query.

19.3 Request / Response Example

Request

```
POST /api/v1/opportunities HTTP/1.1
Authorization: Bearer <jwt>
Idempotency-Key: 7c2a...e91
Content-Type: application/json

{
  "account_id": "0190-...-a1",
  "name": "Acme Platform Renewal",
  "amount": 120000,
  "currency": "USD",
  "stage": "qualification"
}
```

Response (201 Created)

```

{
  "id": "0192-...-7f",
  "account_id": "0190-...-a1",
  "name": "Acme Platform Renewal",
  "amount": 120000,
  "currency": "USD",
  "stage": "qualification",
  "ai_score": 87,
  "created_at": "2026-06-27T10:15:00Z"
}

```

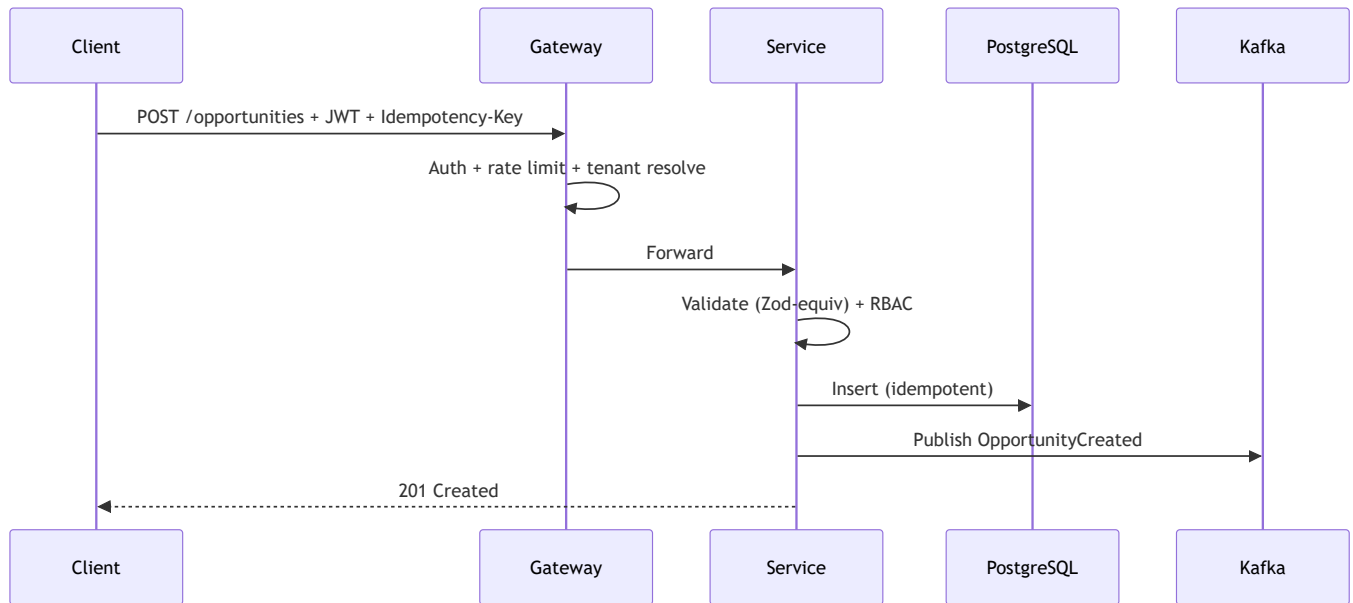
Error (422)

```

{
  "type": "https://horizon360.dev/errors/validation",
  "title": "Validation failed",
  "status": 422,
  "errors": { "amount": ["must be greater than 0"] }
}

```

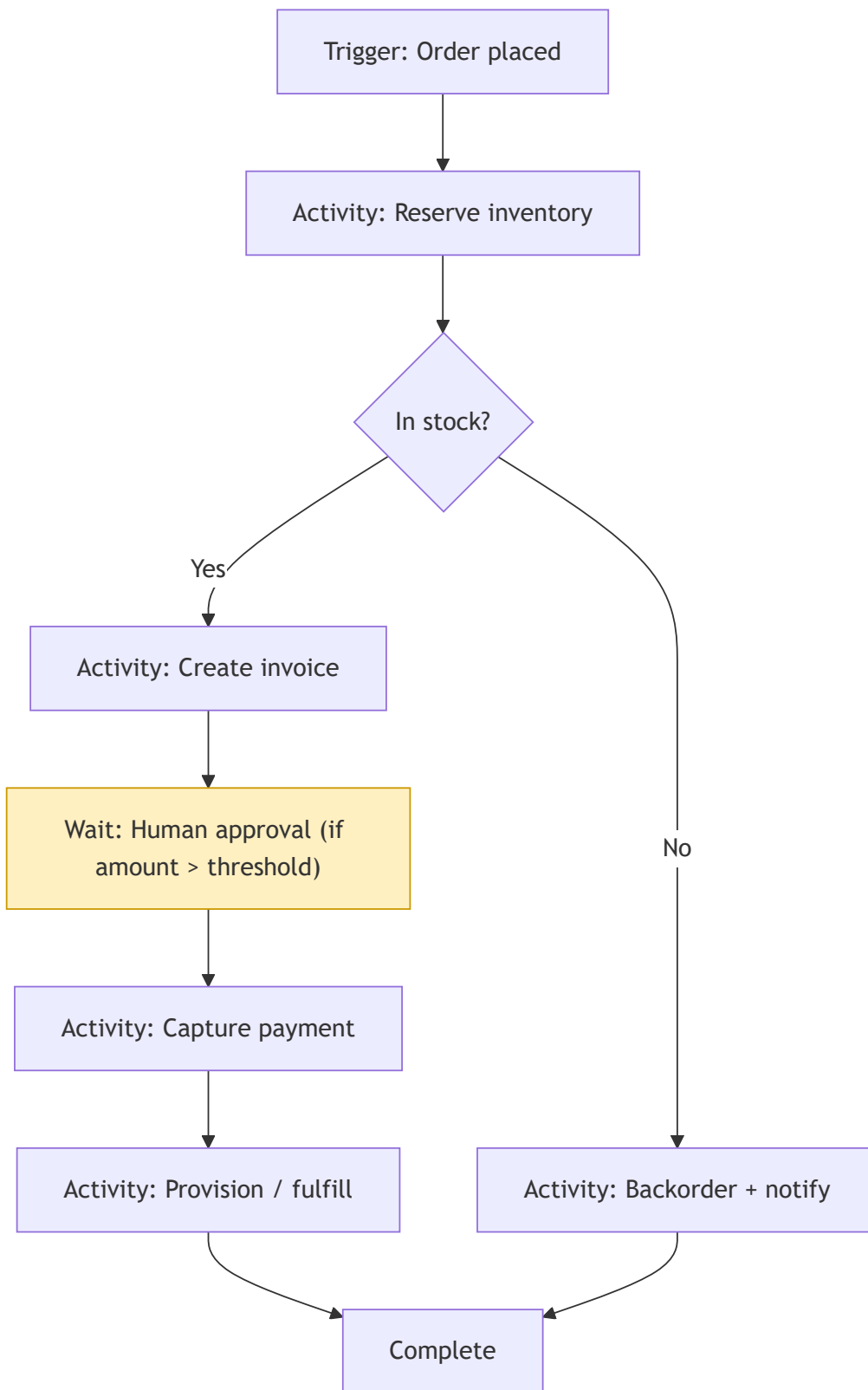
19.4 API Request Lifecycle



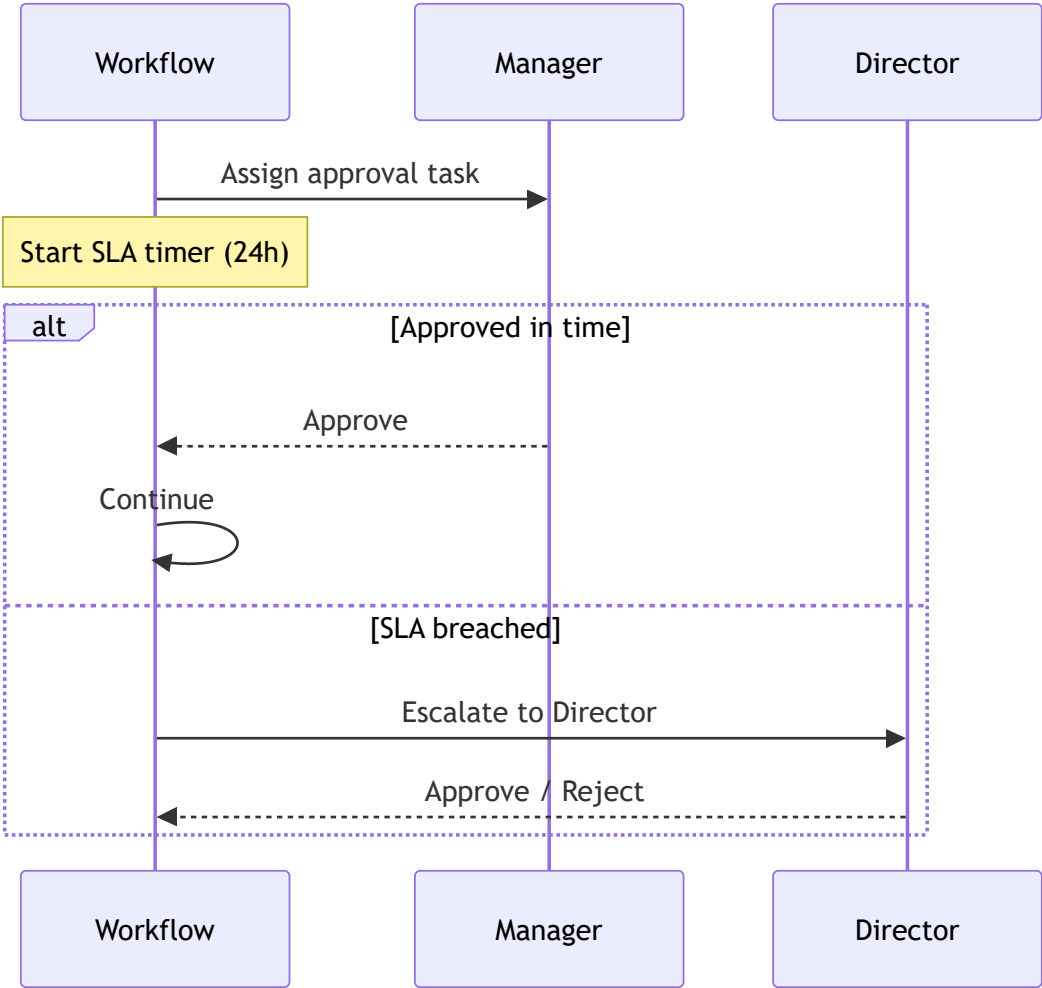
20. Workflow Engine

20.1 Durable Orchestration with Temporal

Business processes are long-running, failure-prone, and span multiple BIOMs. Temporal guarantees that a started workflow runs to completion — surviving crashes, restarts, and timeouts — with full execution history.



20.2 Approval Workflow with Escalation

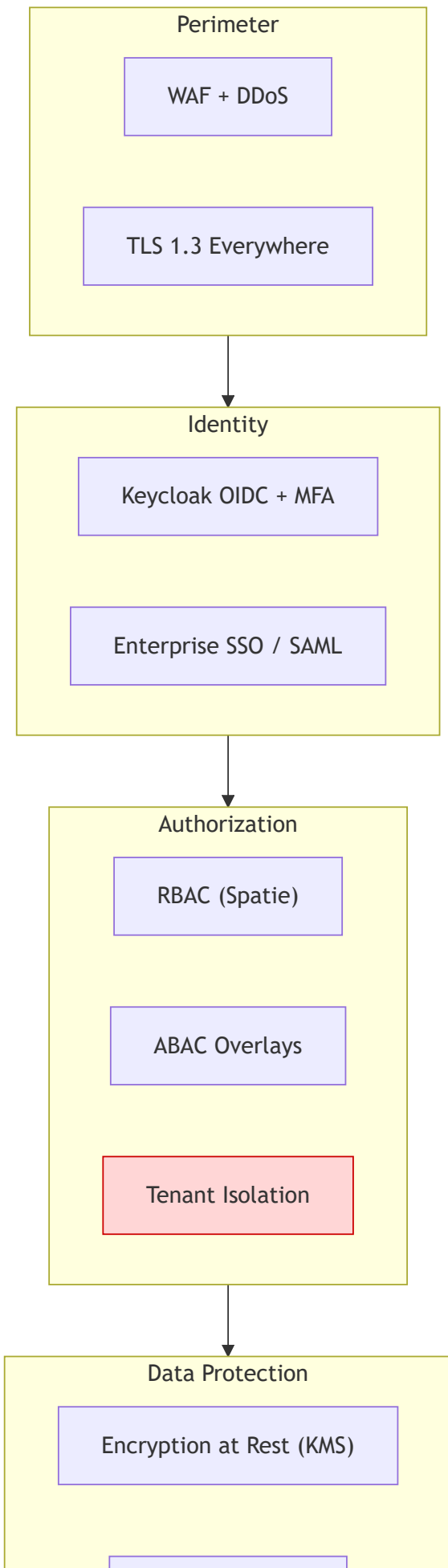


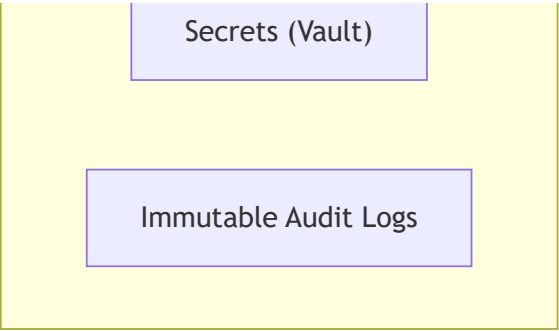
20.3 Engine Capabilities

Capability	Description
Visual Builder	Business users compose workflows from triggers, conditions, actions.
Human Tasks	Workflows pause for approvals, assignments, and reviews.
Retry & Compensation	Automatic retries with backoff; saga-style compensation on failure.
Escalation	Time-bound SLAs with automatic escalation paths.
Auditability	Every workflow execution is fully recorded and replayable.

21. Security Architecture

21.1 Layered Security Model

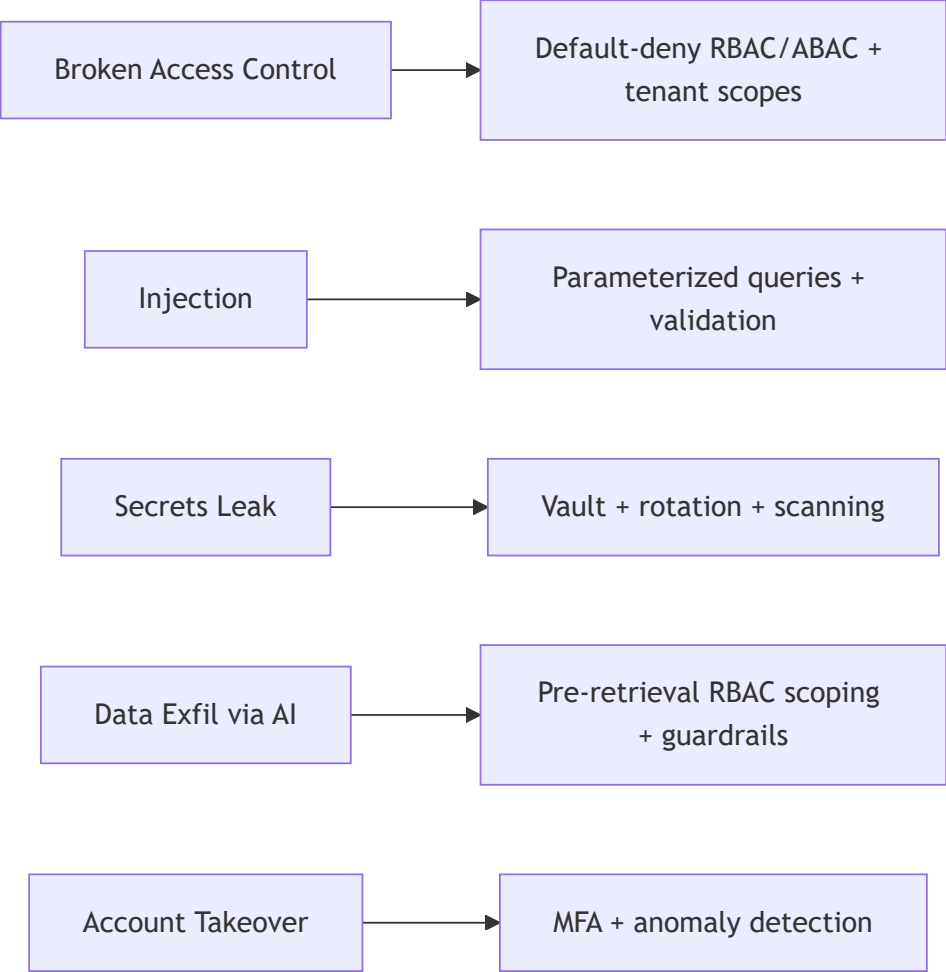




21.2 Controls Matrix

Domain	Control
Authentication	OIDC, MFA, SSO/SAML, short-lived JWTs, refresh rotation.
Authorization	RBAC roles + ABAC attributes; least-privilege defaults.
Encryption	TLS 1.3 in transit; AES-256 at rest via KMS-managed keys.
Secrets	HashiCorp Vault; no secrets in code or env files.
Audit	Append-only, tamper-evident logs; per-tenant retention.
Compliance	SOC2 Type II controls, GDPR (DSAR, erasure, residency).
AppSec	OWASP Top 10 coverage; SAST, DAST, dependency & container scanning.

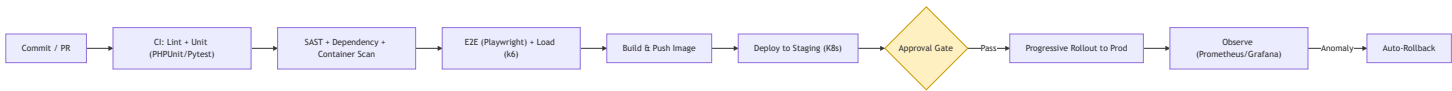
21.3 Threat → Mitigation



⚠ Important Highlight: In an AI-native platform, the AI layer is an attack surface. Horizon 360 treats every AI retrieval as a privileged data access subject to the same RBAC, tenant isolation, and audit controls as a direct API call.

22. DevOps

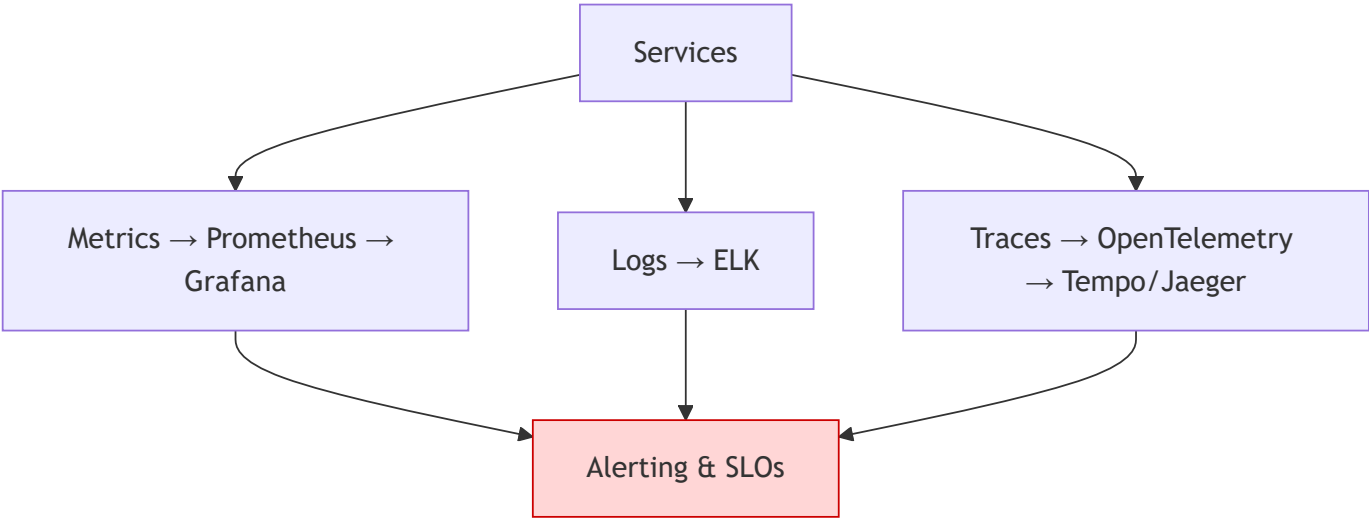
22.1 CI/CD Pipeline



22.2 Infrastructure & Operations

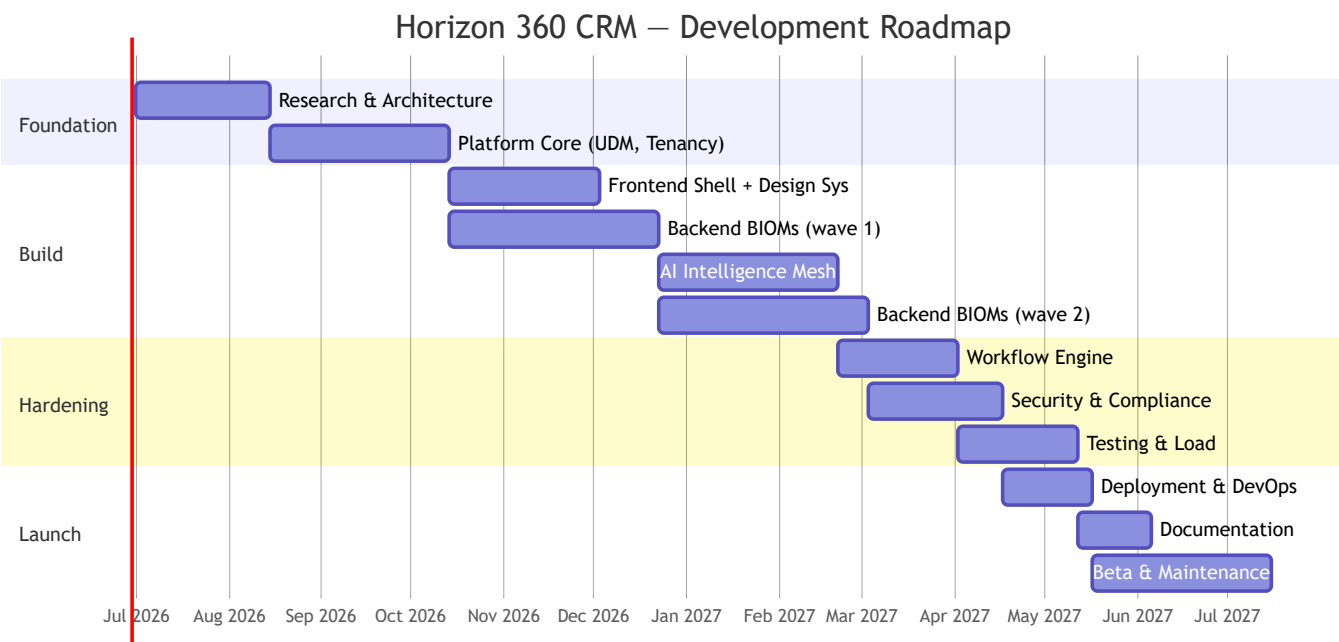
Area	Approach
Infrastructure as Code	Terraform provisions all cloud resources declaratively.
Containers	Docker images per service; immutable, versioned.
Orchestration	Kubernetes with HPA, rolling/canary deploys.
Secrets	Vault injected at runtime.
Monitoring	Prometheus (metrics), Grafana (dashboards), ELK (logs), OpenTelemetry (traces).
Release Strategy	Trunk-based + feature flags; canary → progressive → full.
DR	Multi-AZ, automated backups, tested restore runbooks.

22.3 Observability Pillars



23. Development Roadmap

23.1 Gantt Plan



23.2 Phase Gates

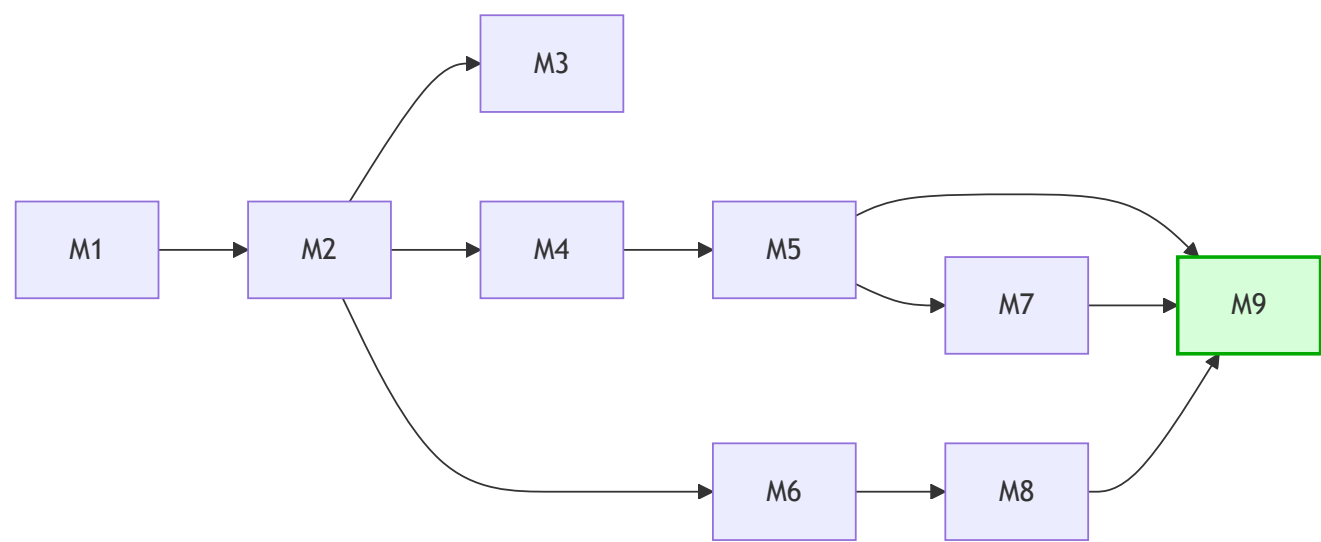
Phase	Exit Criteria
Research	Approved architecture + ADRs.
Core	UDM + tenancy + auth proven at load.
BIOM Wave 1	Sales, Finance, Service live in staging.
AI Mesh	Grounded Copilot with guardrails passing red-team.
Hardening	SOC2 controls in place; load targets met.
Launch	Beta tenants onboarded; runbooks signed off.

24. Project Timeline

24.1 Milestones & Deliverables

Milestone	Target	Deliverable	Depends On
M1 — Architecture Baseline	Month 2	ADRs, UDM schema	—
M2 — Platform Core	Month 4	Tenancy, auth, gateway	M1
M3 — Design System + Shell	Month 6	Component library, app shell	M2
M4 — BIOM Wave 1	Month 7	Sales, Finance, Service	M2
M5 — AI Mesh	Month 9	Copilot + agents	M4
M6 — BIOM Wave 2	Month 9	Marketing, HRMS, Commerce, Project, Vendor, Partner	M2
M7 — Workflow Engine	Month 10	Temporal workflows	M5
M8 — Security & Compliance	Month 11	SOC2 readiness	M6
M9 — GA Launch	Month 12	Production platform	M7, M8

24.2 Dependency Graph



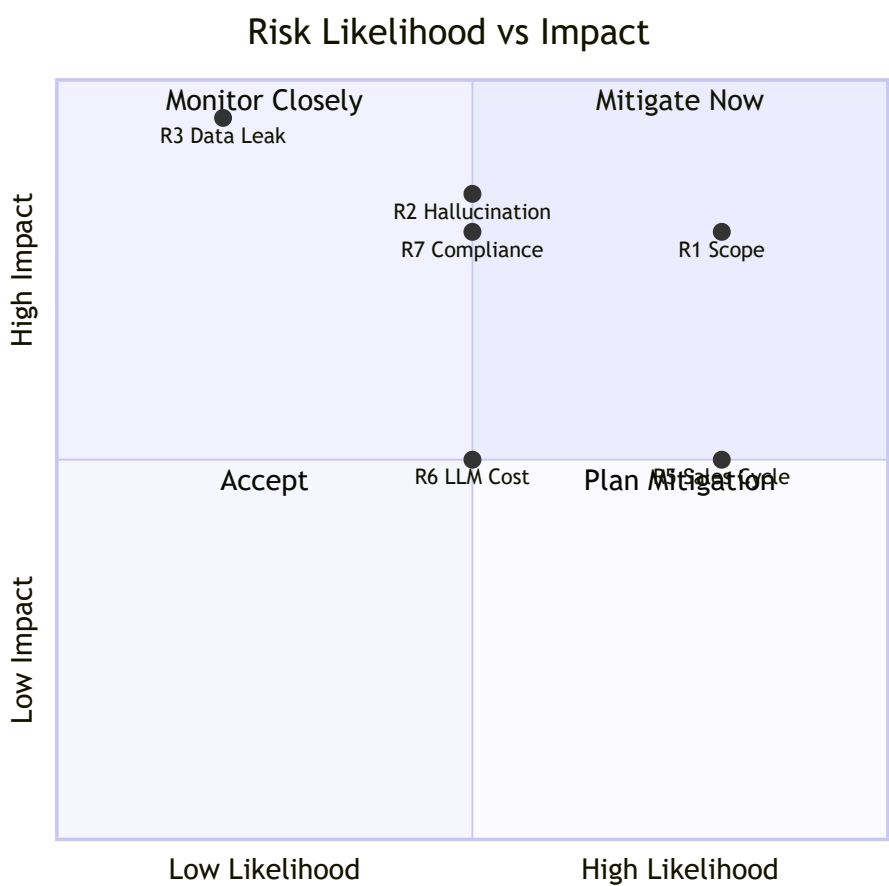
25. Risk Analysis

25.1 Risk Register

ID	Risk	Category	Likelihood	Impact	Mitigation
R1	Scope explosion across 9 BIOMs	Operational	High	High	Phased waves; strict module boundaries; MVP-per-BIOM.
R2	AI hallucination erodes trust	AI	Medium	High	Mandatory grounding, citations, guardrails, human-in-loop.
R3	Multi-tenant data leakage	Security	Low	Critical	Default-deny scopes, RBAC at retrieval, pen tests.
R4	Polyglot ops complexity	Technical	Medium	Medium	Strong IaC, observability, on-call runbooks.
R5	Slow enterprise sales cycle	Business	High	Medium	Land-and-expand; mid-market first; free tier.
R6	LLM cost volatility	Business	Medium	Medium	Model routing, caching, in-house ONNX for hot paths.
R7	Compliance gaps block deals	Operational	Medium	High	Compliance-by-design; SOC2 early.

ID	Risk	Category	Likelihood	Impact	Mitigation
R8	Workflow failures lose processes	Technical	Low	High	Temporal durable execution; idempotency.

25.2 Risk Heat Map



26. Competitor Analysis

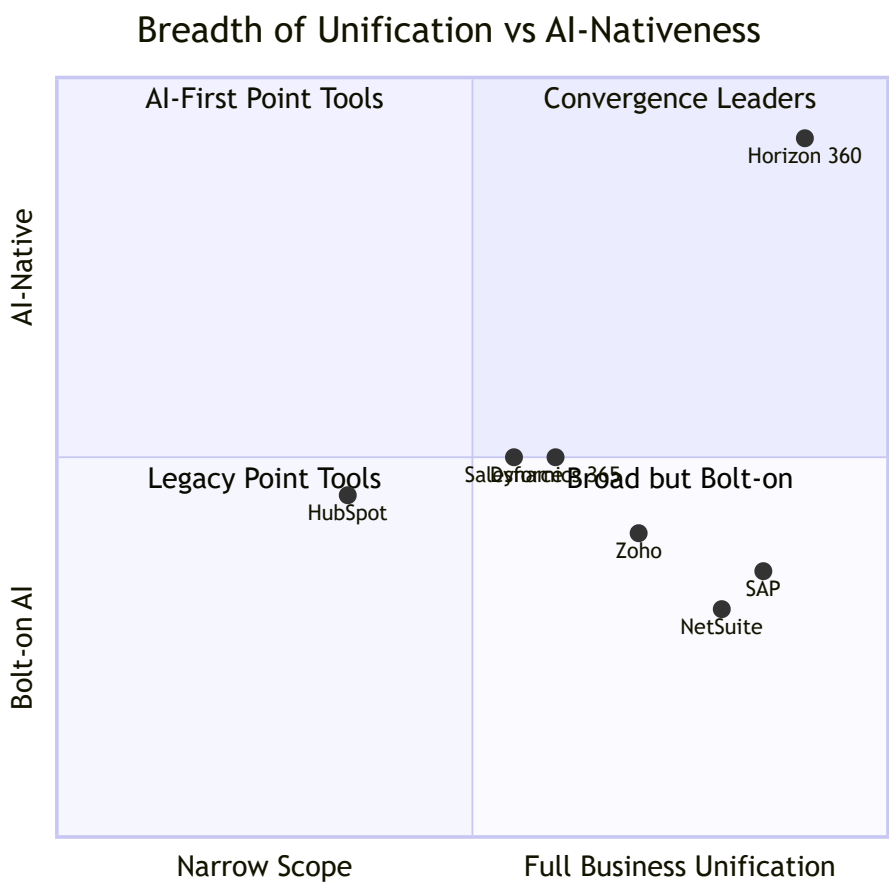
26.1 Feature Comparison

Capability	Horizon 360	Salesforce	HubSpot	Zoho	MS Dynamics 365	SAP	Oracle NetSuite
Unified data model (CRM+ERP+HR)	✓ Native	⚠ Add-ons/clouds	⚠ CRM-centric	✓ Broad suite	⚠ Multiple apps	✓ ERP-centric	✓ ERP-centric

Capability	Horizon 360	Salesforce	HubSpot	Zoho	MS Dynamics 365	SAP	Oracle NetSuite
AI-native (mesh, shared memory)	✓ Core	⚠ Einstein add-on	⚠ Limited	⚠ Emerging	⚠ Copilot add-on	⚠ Joule add-on	⚠ Limited
Durable workflow engine	✓ Temporal	⚠ Flows	⚠ Workflows	⚠ Basic	⚠ Power Automate	✓ Strong	⚠ SuiteFlow
Single SaaS codebase, composable	✓ BIOMs	✗ Multi-cloud	⚠ Hubs	✓	✗ Apps	✗ Heavy	⚠
Time-to-value	✓ Fast	⚠ Slow	✓ Fast	✓	⚠	✗ Slow	⚠
TCO at convergence	✓ Low	✗ High	⚠ Medium	✓ Low	✗ High	✗ High	⚠ Medium

(Legend: ✓ strong · ⚠ partial · ✗ weak. Assessment is directional positioning, not a benchmarked audit — validate before external use.)

26.2 Strategic Positioning



💡 **Best Practice Note:** Horizon 360 does not win by out-featuring Salesforce in CRM or SAP in ERP. It wins on the axis neither can easily move on: a *single* AI-native unified model. Competing on the incumbent's home turf is a losing strategy; redefining the axis is the play.

27. Business Feasibility

27.1 Market & Pain

The convergence thesis is validated by a clear, recurring customer pain: vendor sprawl. Mid-market firms (200–2,000 employees) are the sweet spot — large enough to feel fragmentation pain, small enough to consolidate without multi-year migrations.

27.2 Feasibility Dimensions

Dimension	Assessment
Market Size	Large and growing across CRM/ERP/HR/BI convergence.

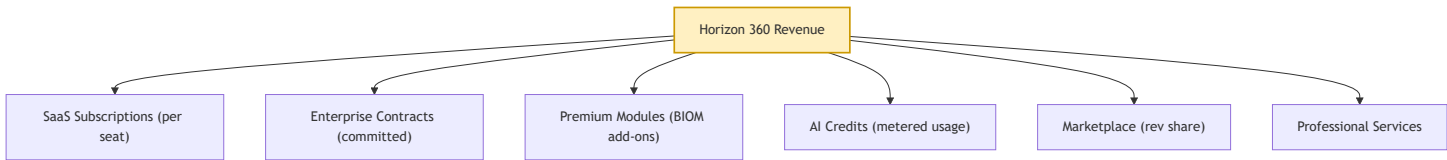
Dimension	Assessment
Customer Pain	Acute, recurring, budget-backed (consolidation initiatives).
Revenue Opportunity	Multi-product expansion per account (high NRR potential).
Scalability	Multi-tenant SaaS — marginal cost per tenant approaches zero.
Growth Strategy	Land with one BIOM, expand across the estate.

27.3 Land-and-Expand Motion



28. Revenue Model

28.1 Streams



28.2 Packaging

Tier	Audience	Includes	Monetization
Starter	Small business	Core CRM + 1 BIOM	Per-seat, monthly
Growth	Mid-market	Core + 4 BIOMs + AI base	Per-seat + AI credits
Enterprise	Large org	All BIOMs, SSO, dedicated tenancy	Committed annual + services
Platform	ISVs/Partners	API + marketplace	Usage + revenue share

🔴

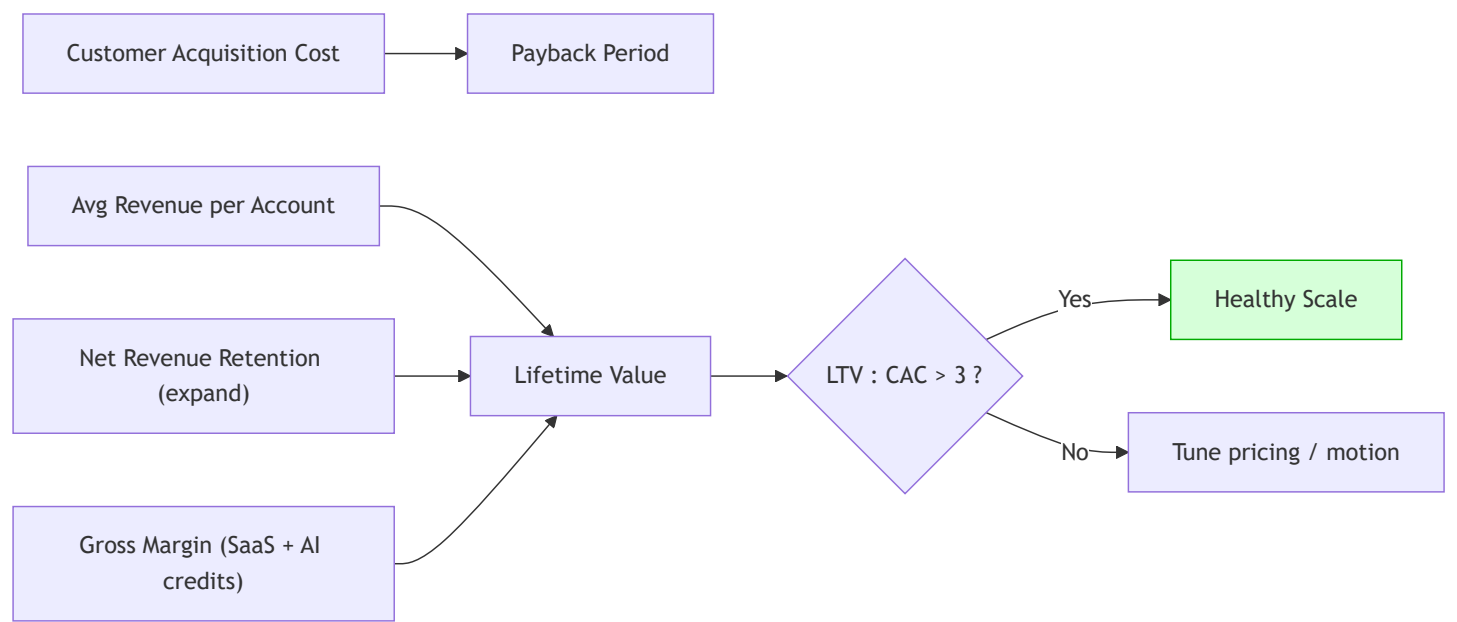
Important Highlight: AI Credits decouple AI cost from seat price. Customers pay for the intelligence they consume, protecting gross margin against LLM cost volatility (mitigating risk R6).

29. Investment Analysis

29.1 Cost Structure (Indicative Model)

Category	Description	Relative Weight
Development	Engineering, product, design across 12-month build.	High
AI Costs	Training, inference, LLM API/usage.	Medium-High
Cloud Infrastructure	K8s, databases, storage, networking.	Medium
Security & Compliance	SOC2, pen tests, tooling.	Medium
Marketing	Brand, demand gen, content.	Medium
Operations	Support, SRE, G&A.	Medium

29.2 Unit Economics Logic



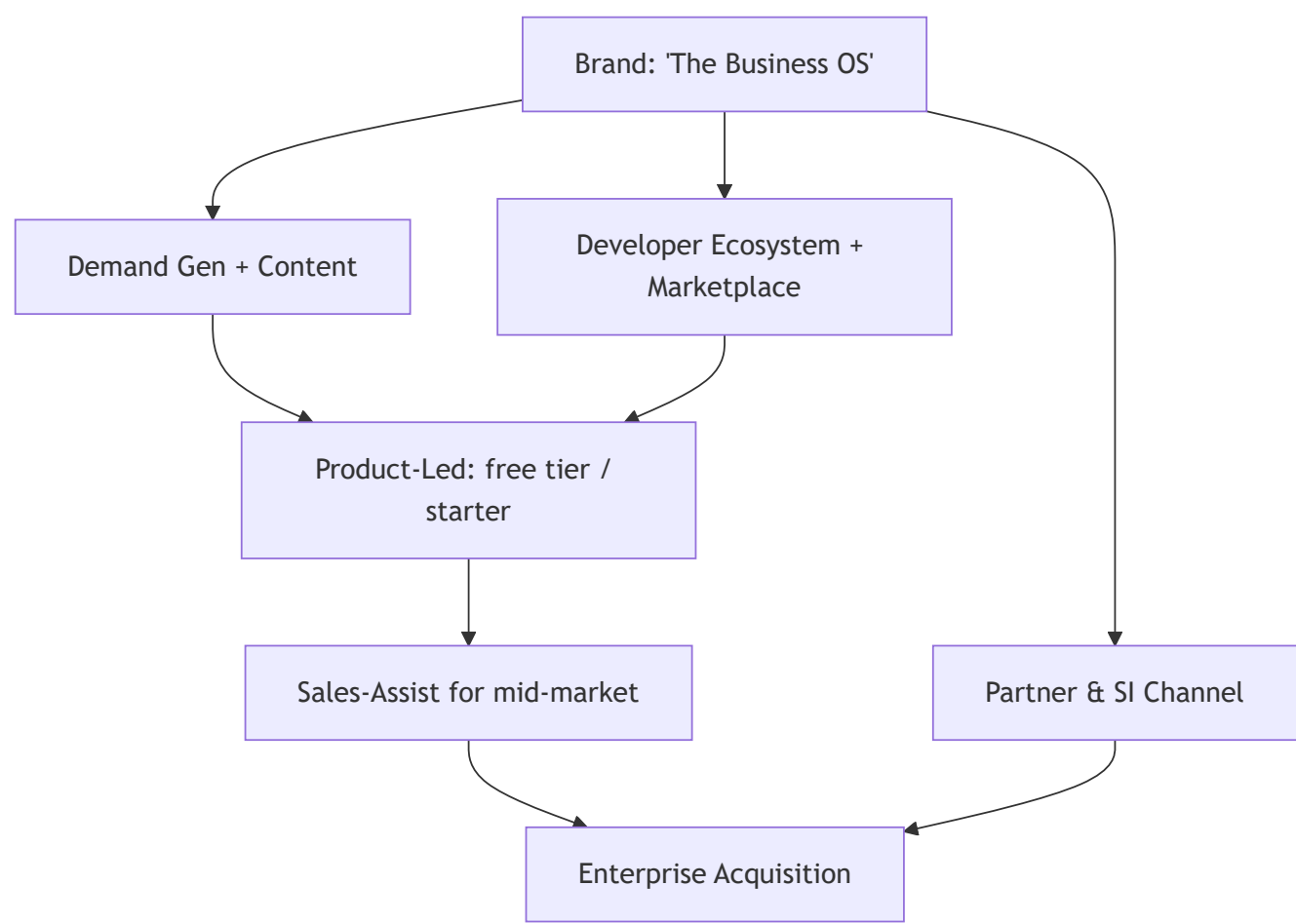
29.3 Break-Even & ROI

Break-even is reached when recurring gross profit covers fixed operating cost. The land-and-expand motion drives **Net Revenue Retention above 100%**, meaning the existing base grows revenue even before new logos — the structural advantage of a unified platform with many expansion vectors.

⚠ Best Practice Note: All figures in this chapter are a *modeling framework*, not committed financials. Before any investor discussion, populate with bottoms-up numbers and validate against comparable public SaaS

30. Go-To-Market Strategy

30.1 GTM Engine

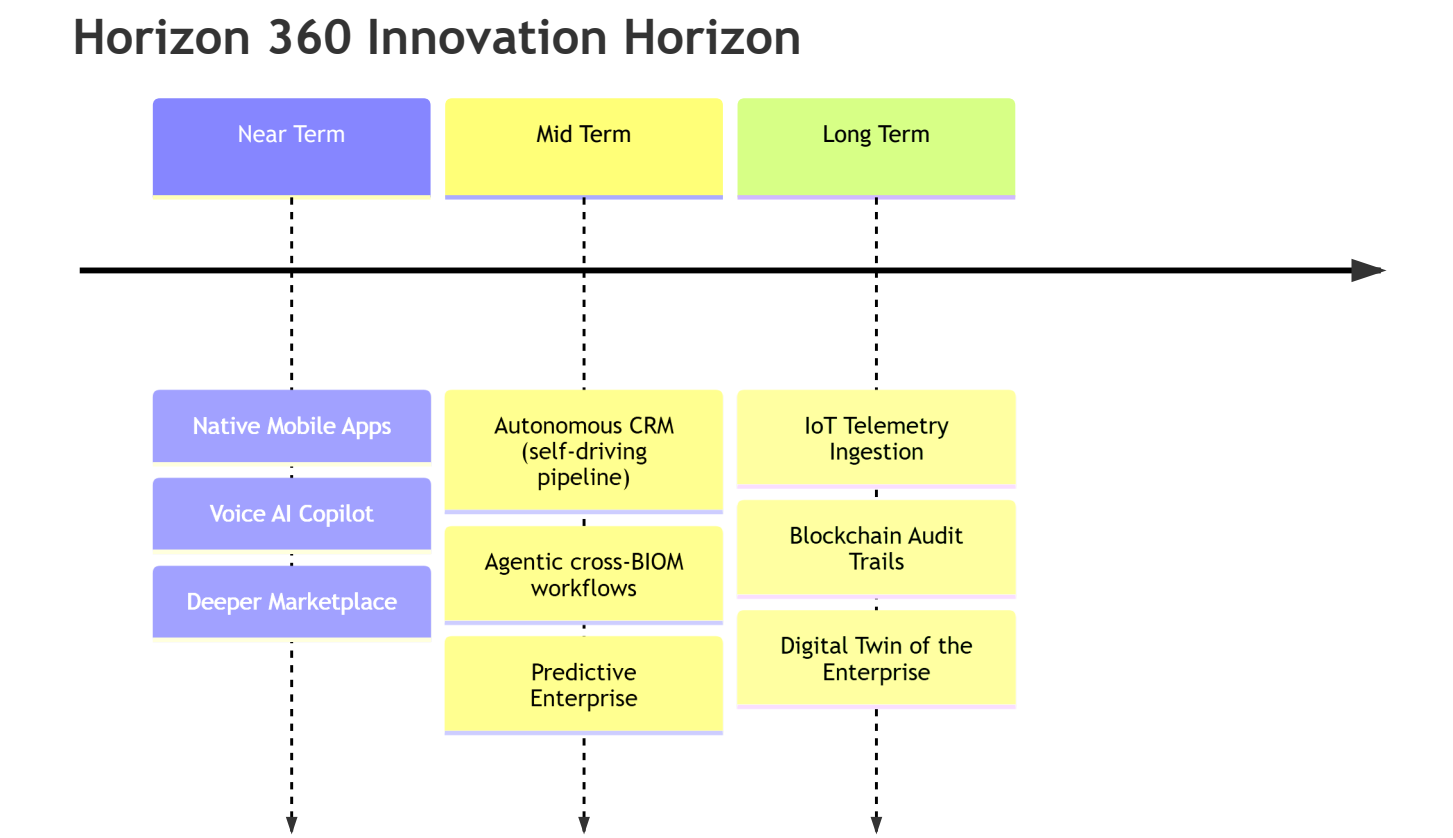


30.2 Channel Plan

Channel	Motion
Branding	Own the "Business Operating System" category narrative.
Sales	Product-led for SMB; sales-assisted for mid-market; field for enterprise.
Marketing	Content on consolidation ROI; comparison vs. multi-vendor stacks.
Enterprise Acquisition	ABM on consolidation-minded accounts; proof-of-value pilots.

Channel	Motion
Partner Strategy	SIs and consultancies for implementation leverage.
Community	Practitioner community around the platform.
Developer Ecosystem	Open APIs + marketplace to drive stickiness and reach.

31. Future Enhancements



Enhancement	Vision
AI Agents	From assistive to agentic — agents that execute multi-step business processes autonomously, under guardrails.
Autonomous CRM	The system proactively advances pipeline and flags risk without prompting.
Voice AI	Hands-free Copilot for executives and field teams.
Mobile Apps	Native iOS/Android beyond responsive web.

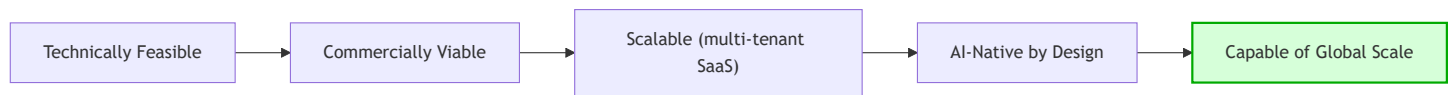
Enhancement	Vision
IoT	Ingest device telemetry into the Universal Data Model.
Blockchain	Tamper-evident audit and settlement where required.
Predictive Enterprise	Shift from descriptive to prescriptive operations.
Digital Twin	A live simulation of the business for scenario planning.

32. Conclusion

Horizon 360 CRM is not an incremental product; it is a **category bet** — that the future of business software is convergence, and that convergence is only achievable at the core, on a single unified data model with native intelligence.

The technical foundation is sound: a proven, modern stack (Next.js, Laravel 12, FastAPI, PostgreSQL, Temporal, Kafka, Qdrant) deliberately composed so each component does what it does best. The architecture is disciplined: an 8-Level Business Operating System, a composable BIOM Ecosystem, an AI Intelligence Mesh scoped by RBAC and tenancy, a durable Workflow Engine, and an Executive Intelligence Center that turns the whole business into a single, queryable surface.

The commercial logic is equally sound: a land-and-expand motion against acute, budget-backed customer pain, with multiple expansion vectors driving net revenue retention above 100%, and AI Credits protecting margin against cost volatility.



 **Final Statement:** Horizon 360 CRM is technically feasible, commercially viable, architecturally scalable, AI-native at its core, and positioned to become a global enterprise platform. It does not ask the market to adopt another tool. It asks the market to adopt an operating system for the business itself.

— End of Project Proposal Documentation —

Horizon 360 CRM · The Universal Business Operating System

Connect Everything · Understand Everything · Automate Everything · Grow Everything